

NIGHT PERSON ALARM

CLASSIFICATION PROBLEM

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Problem definition

Community swimming pool with no nighttime lighting

Recent incident: unauthorized people entered the pool at night

Current solution and its limitations

Install a security camera

But requires a human monitoring → expensive and inefficient

First proposed solution

Binary classification: Person / No person → design our model

Sends automatic ALERT to the community if someone is found

Key challenge

Model must work reliably under low-quality, nighttime conditions

State of the art

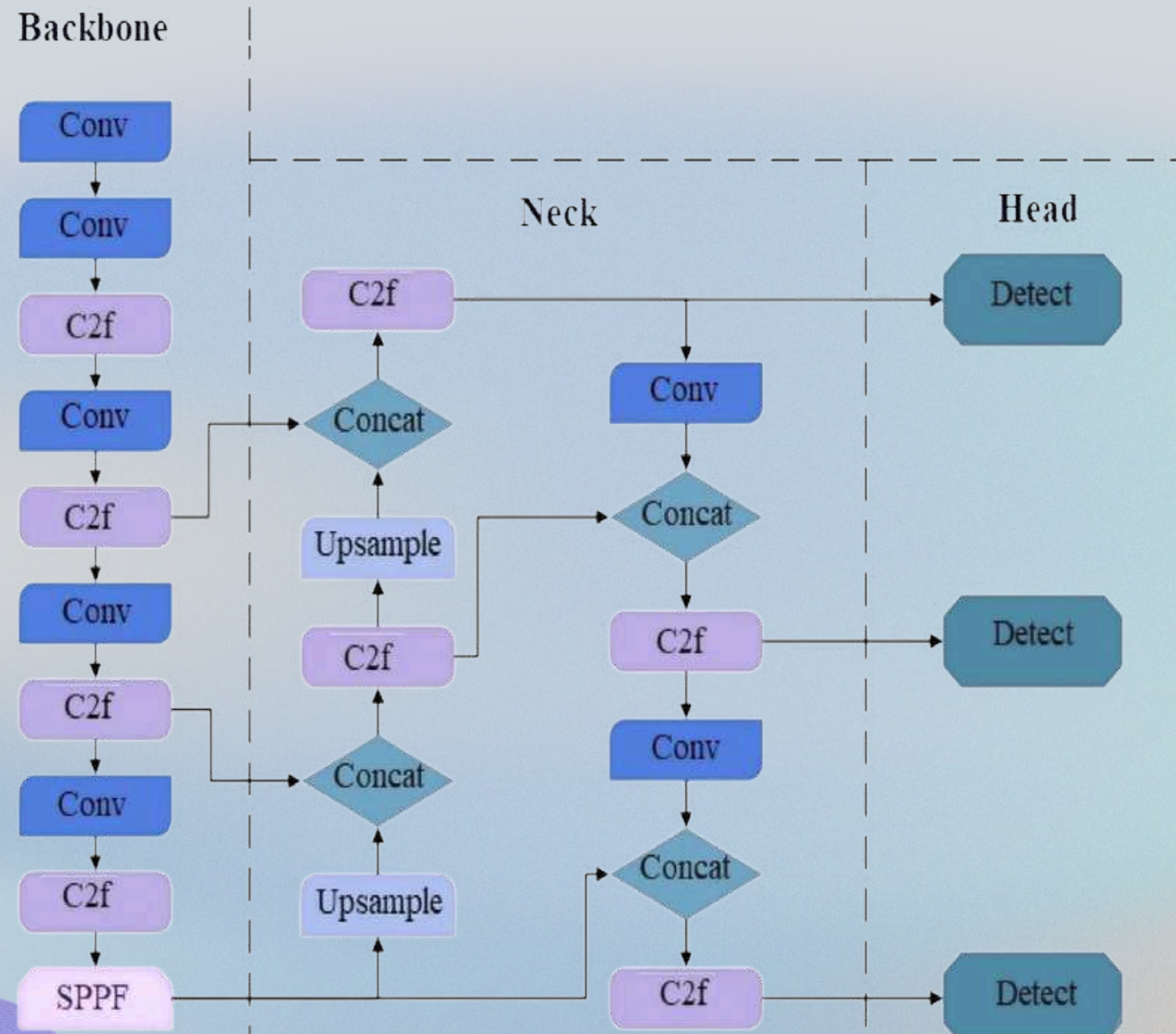
YOLO →

One of the most used architectures for real-time object detection. Localization and classification in a single forward pass.

RTMDet →

Designed to maximize the trade-off between speed and accuracy while maintaining a simple architecture

YOLO Structure





DATASET AND PREPROCESSING

Datasets

LLVIP dataset

Contains infrared images with people: **15488**

divide

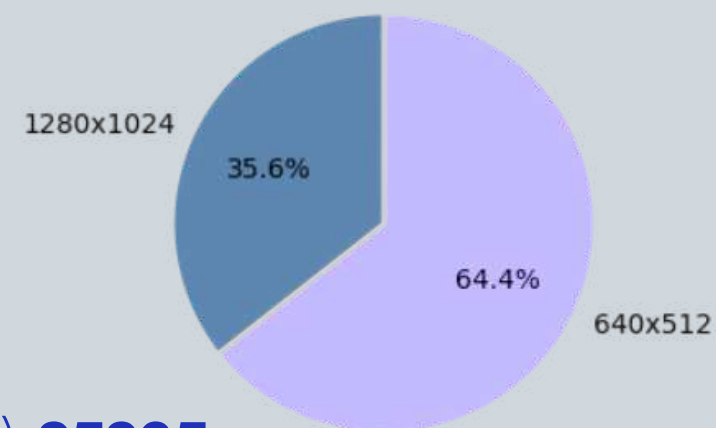
Images divided: **7001**

Images NOT divided: **8487**

Images WITH people (label=1): **25235**

Images WITHOUT people (label=0): **18257**

Generated images: **43492**



My dataset

Collected specifically for this project

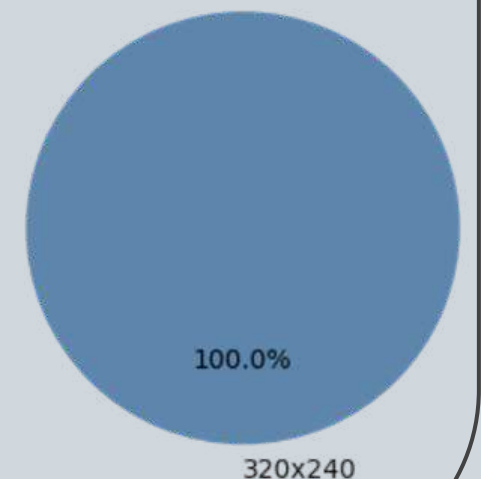
Night grayscale images from the community pool environment

Manually labeled: Person / No Person

Images WITH people (label=1): **978**

Images WITHOUT people (label=0): **355**

Images: **1333**



Why two datasets?

LLVIP provides a large, diverse base for training

Custom dataset ensures the model adapts to our specific scenario

Data Analysis: LLVIP divided

Random Samples

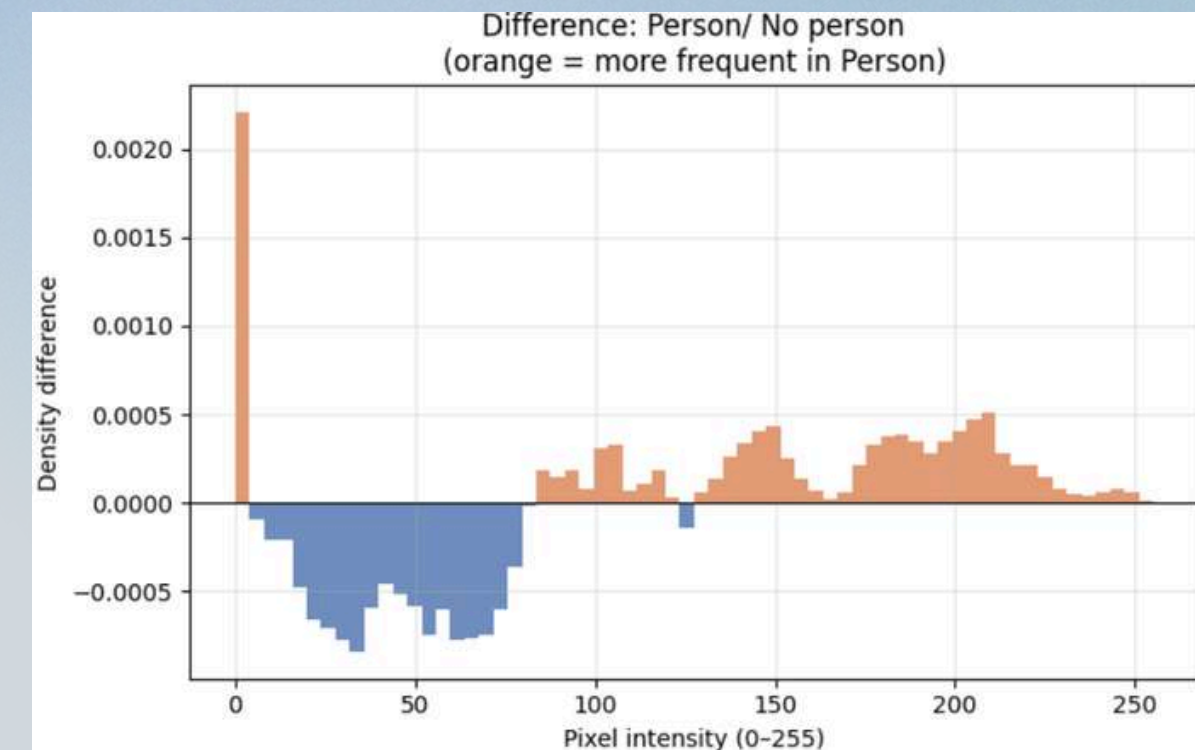
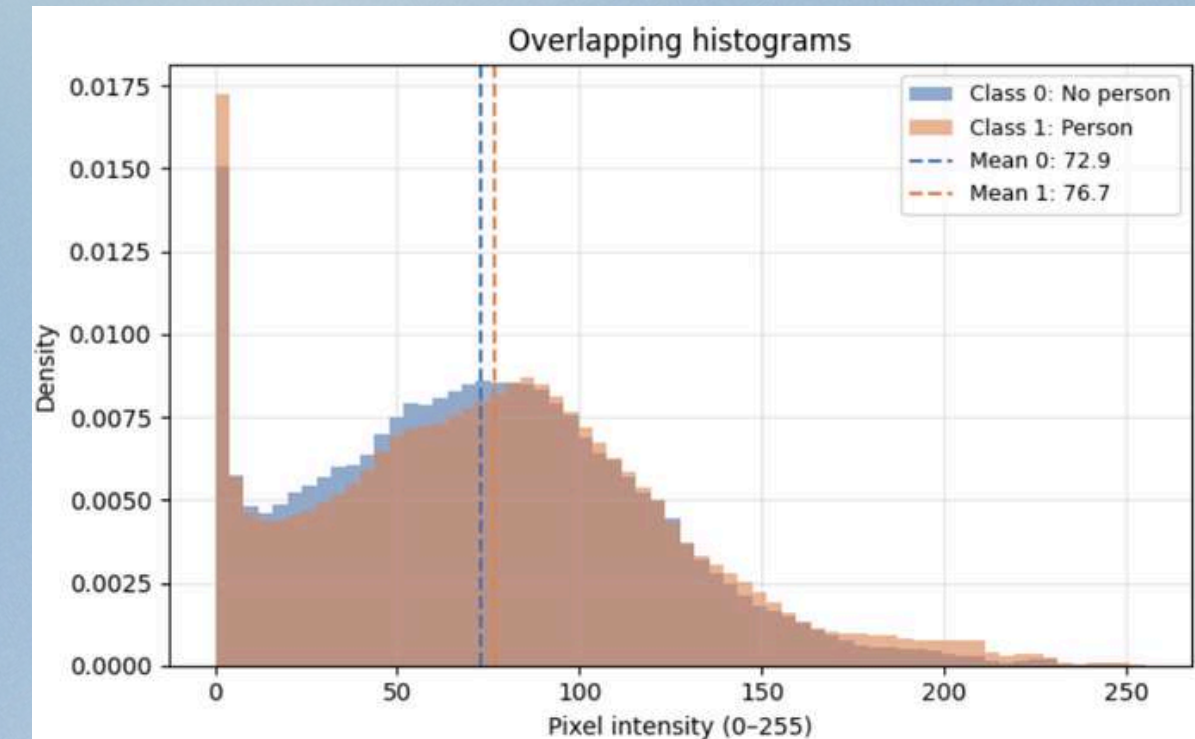
Class 0: No person



Class 1: Person

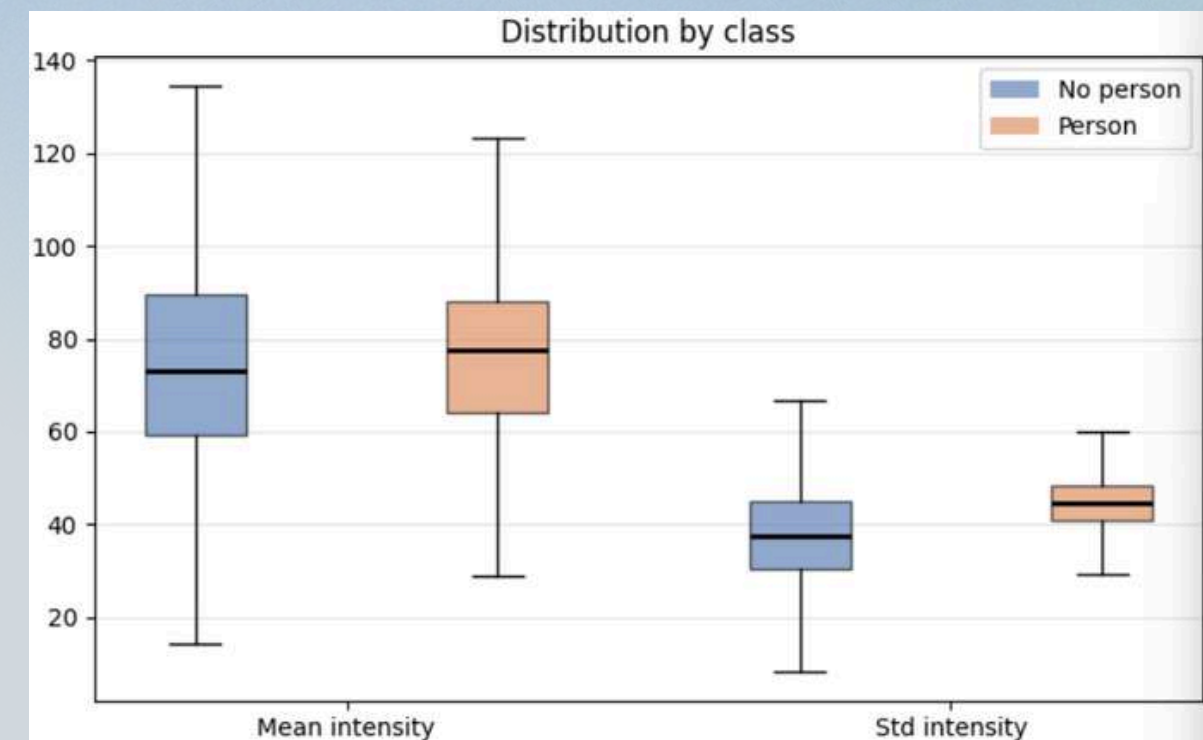
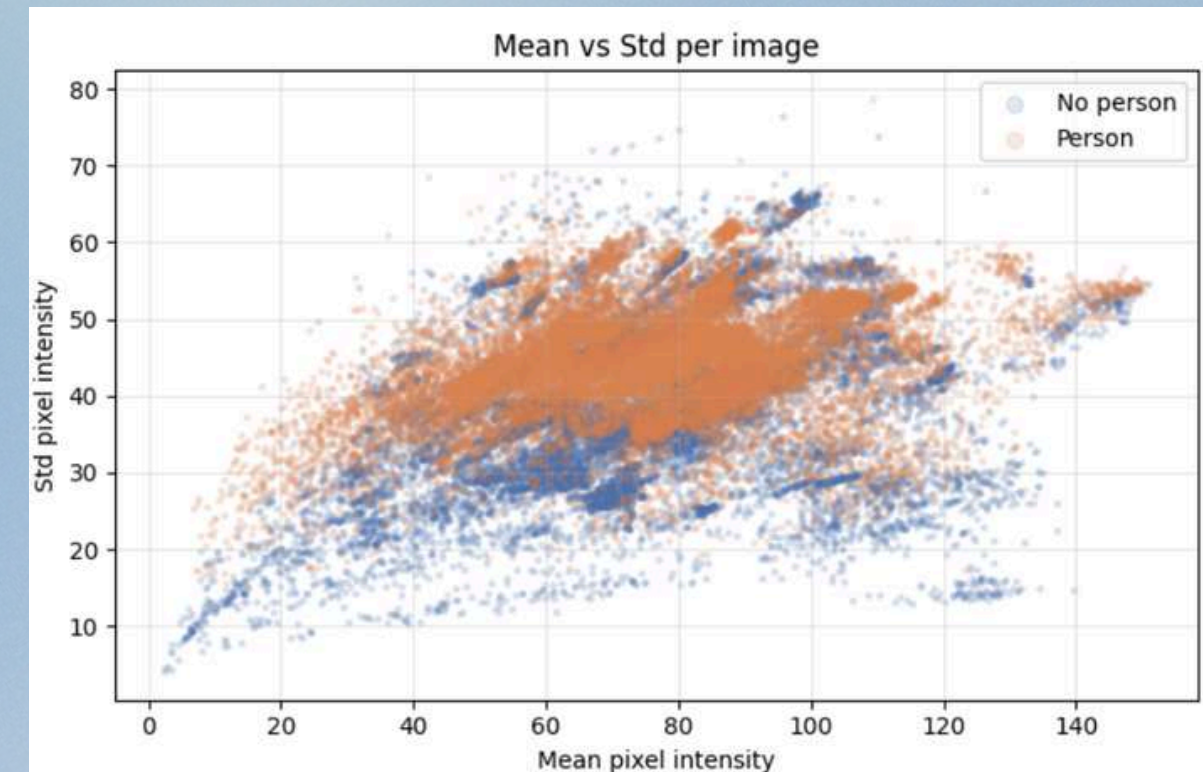
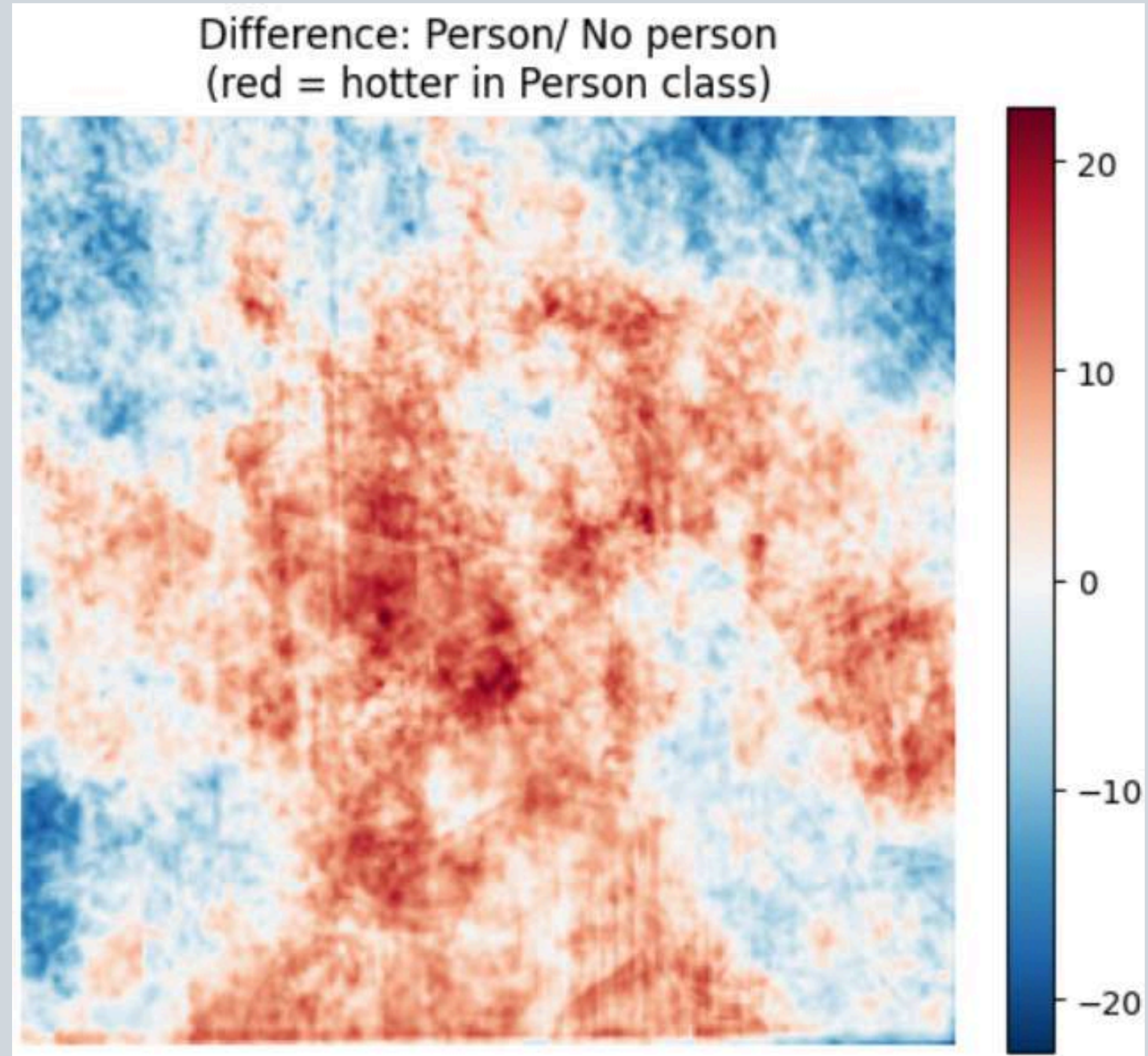


Pixel intensity distribution



Data Analysis: LLVIP divided

Intensity analysis



Data Analysis: My dataset

Random Samples

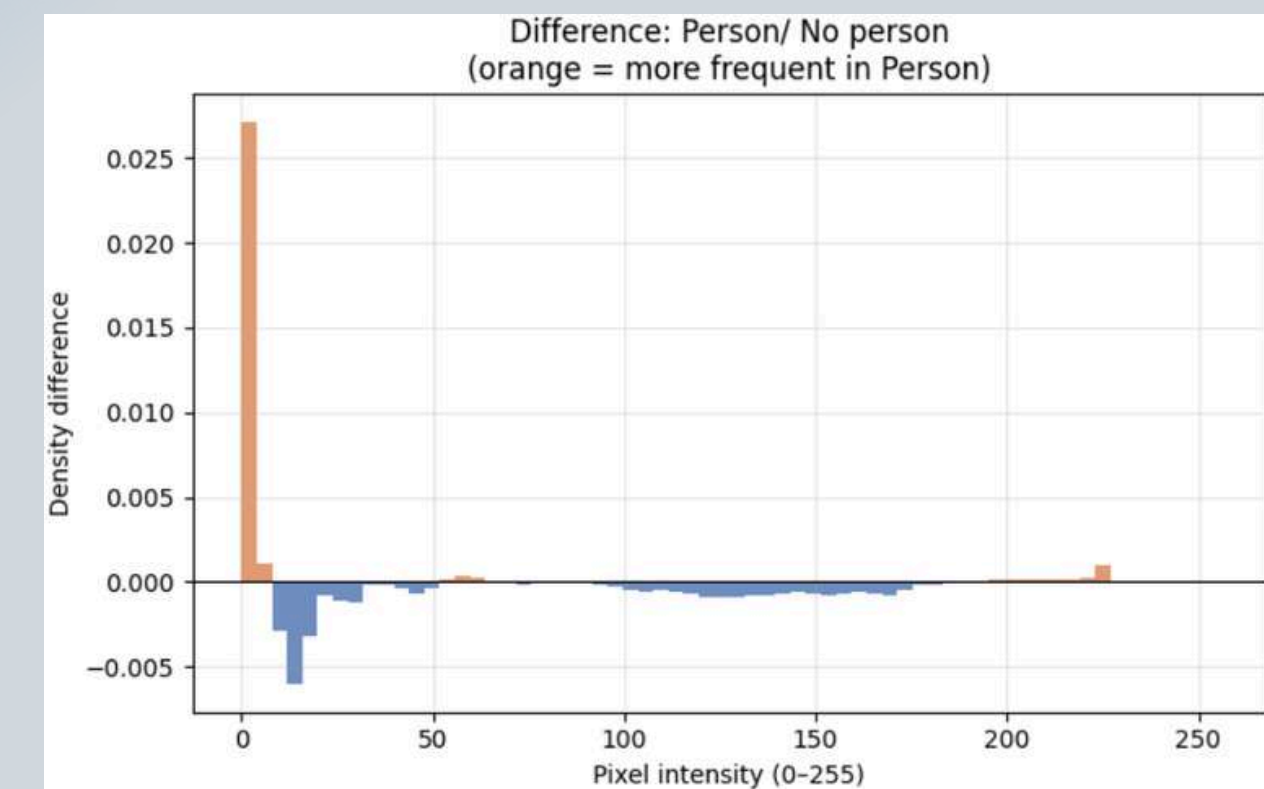
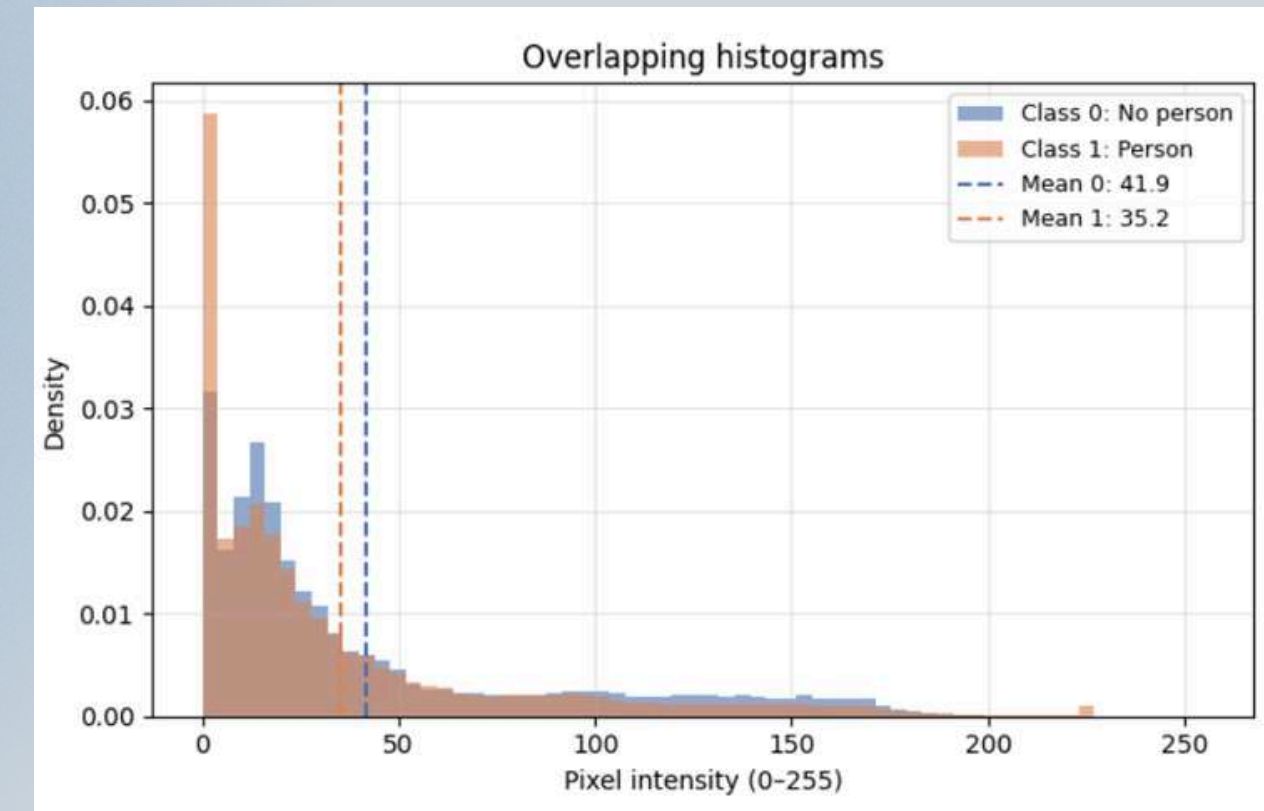
Class 0: No person



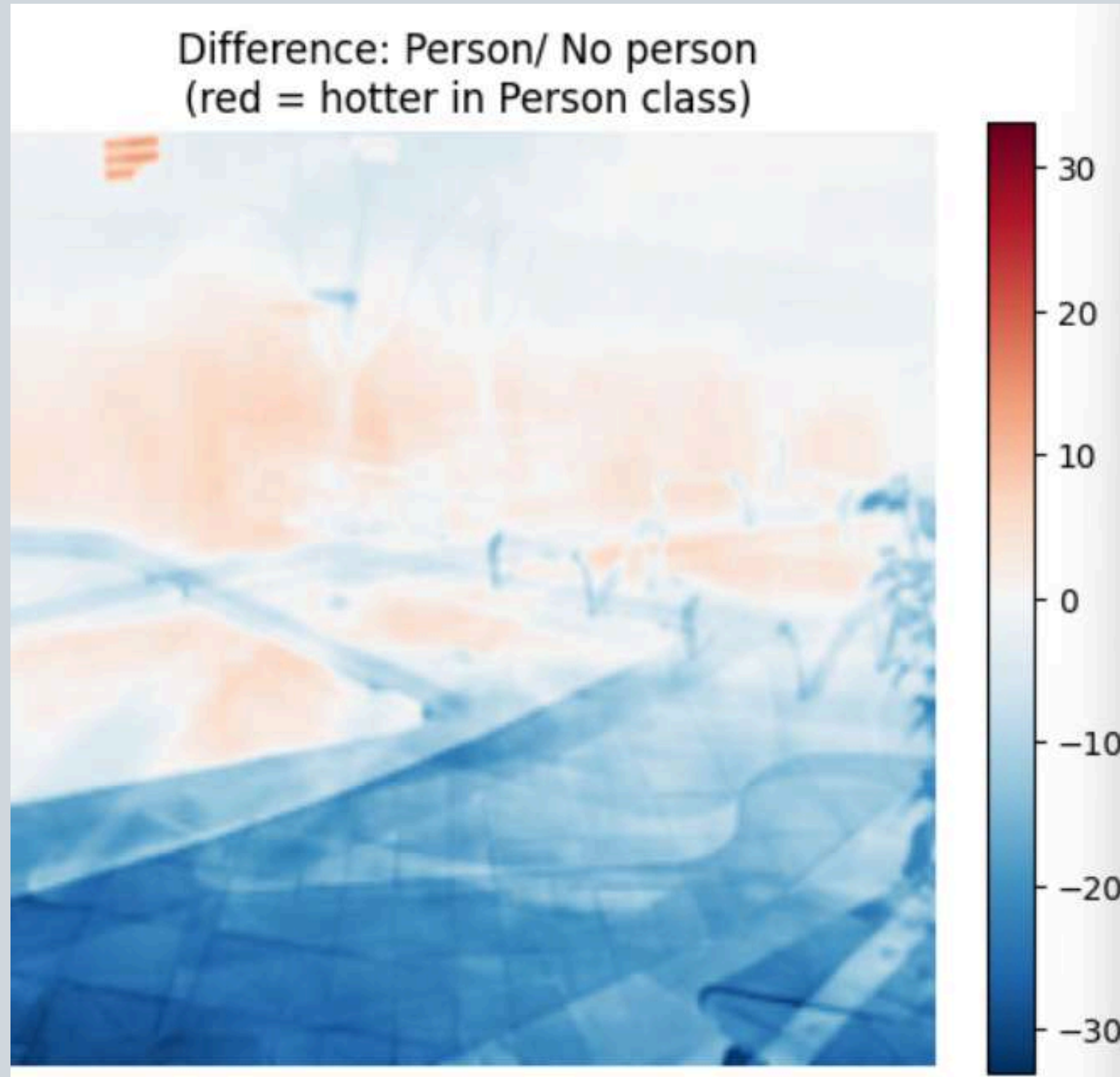
Class 1: Person



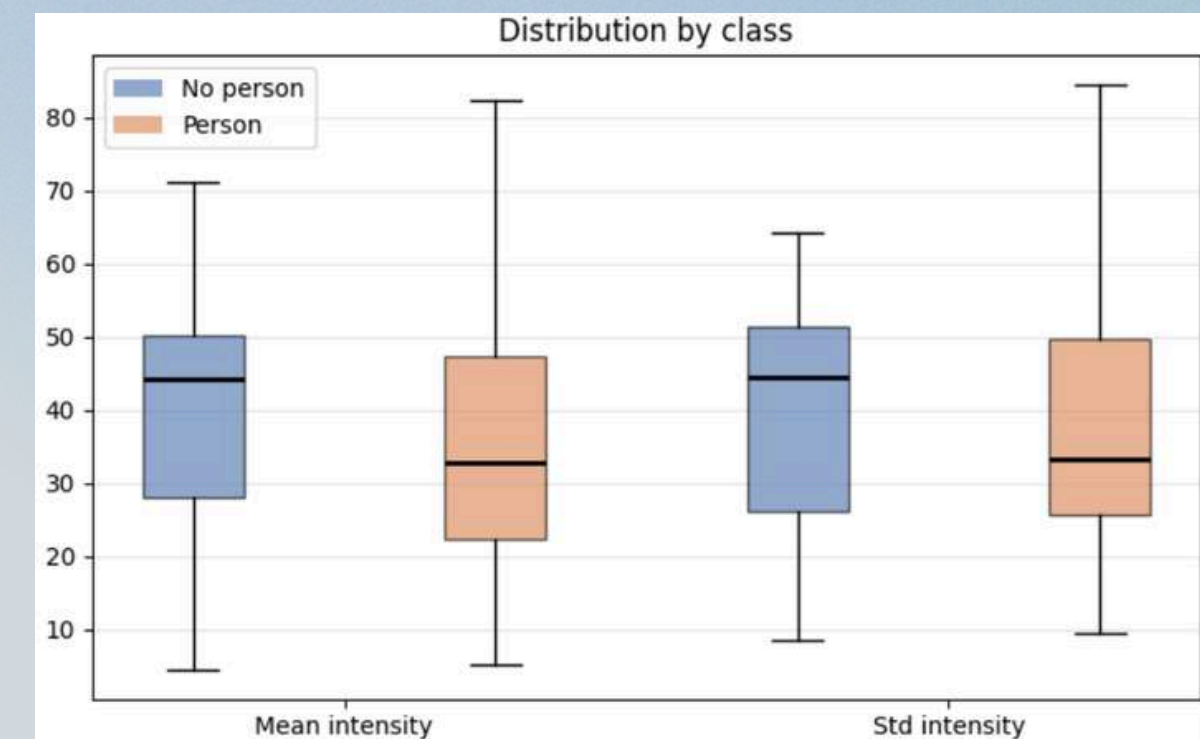
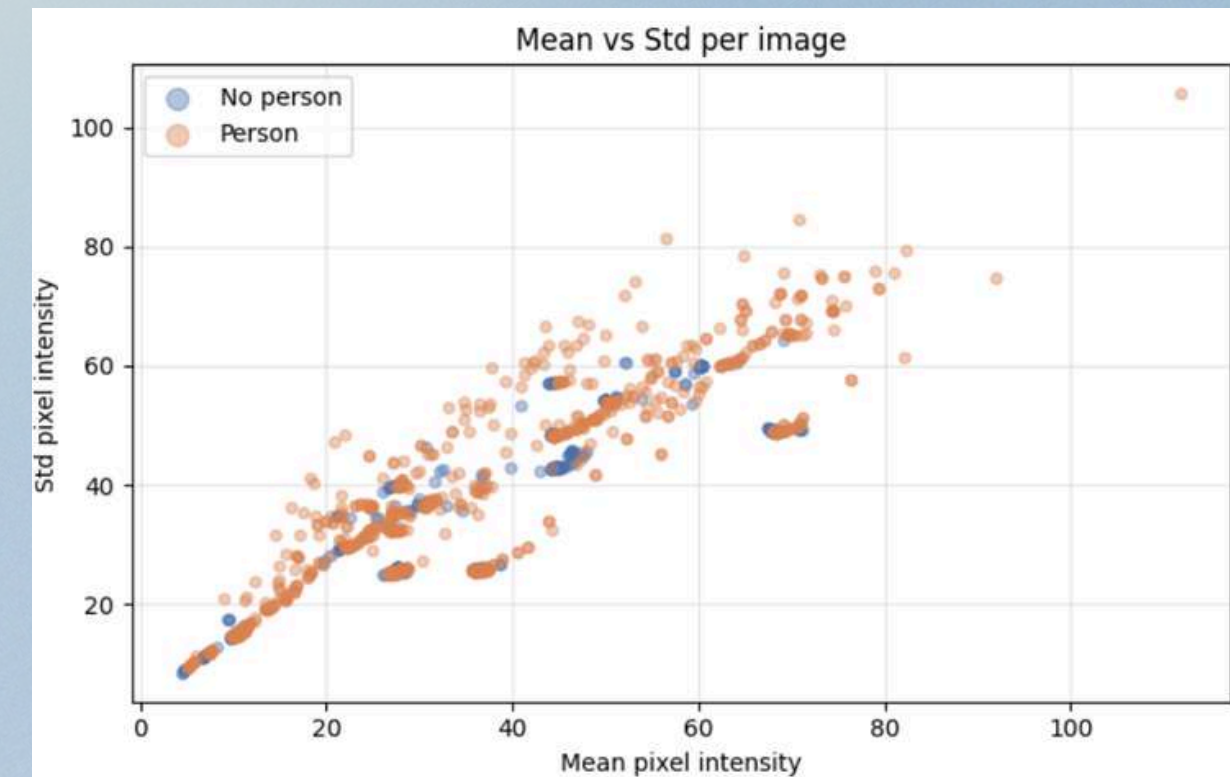
Pixel intensity distribution



Data Analysis: My dataset



Intensity analysis



Difficult cases:

Person class (lowest global contrast)

My dataset



LLVIP



PREPROCESSING

Splits:

LLVIP dataset

LLVIP train (33277) → 80% train, 20 % val
LLVIP test (10215)

My dataset

70% train, 15% val, 15% test

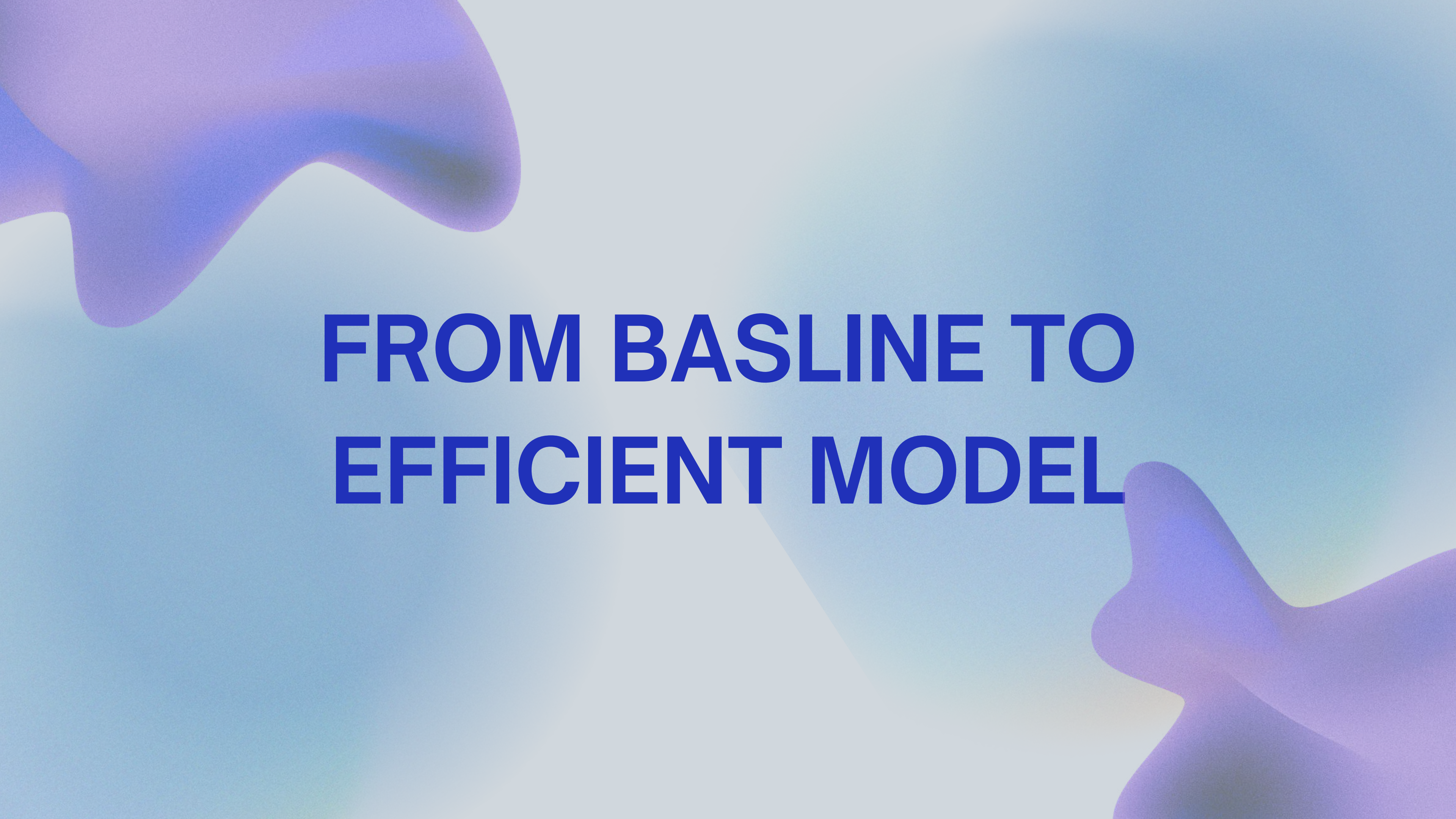
Applied to all splits: train, val, and test

Resize 224×224 → match the input size expected by the model

Converts Image to a **PyTorch tensor**

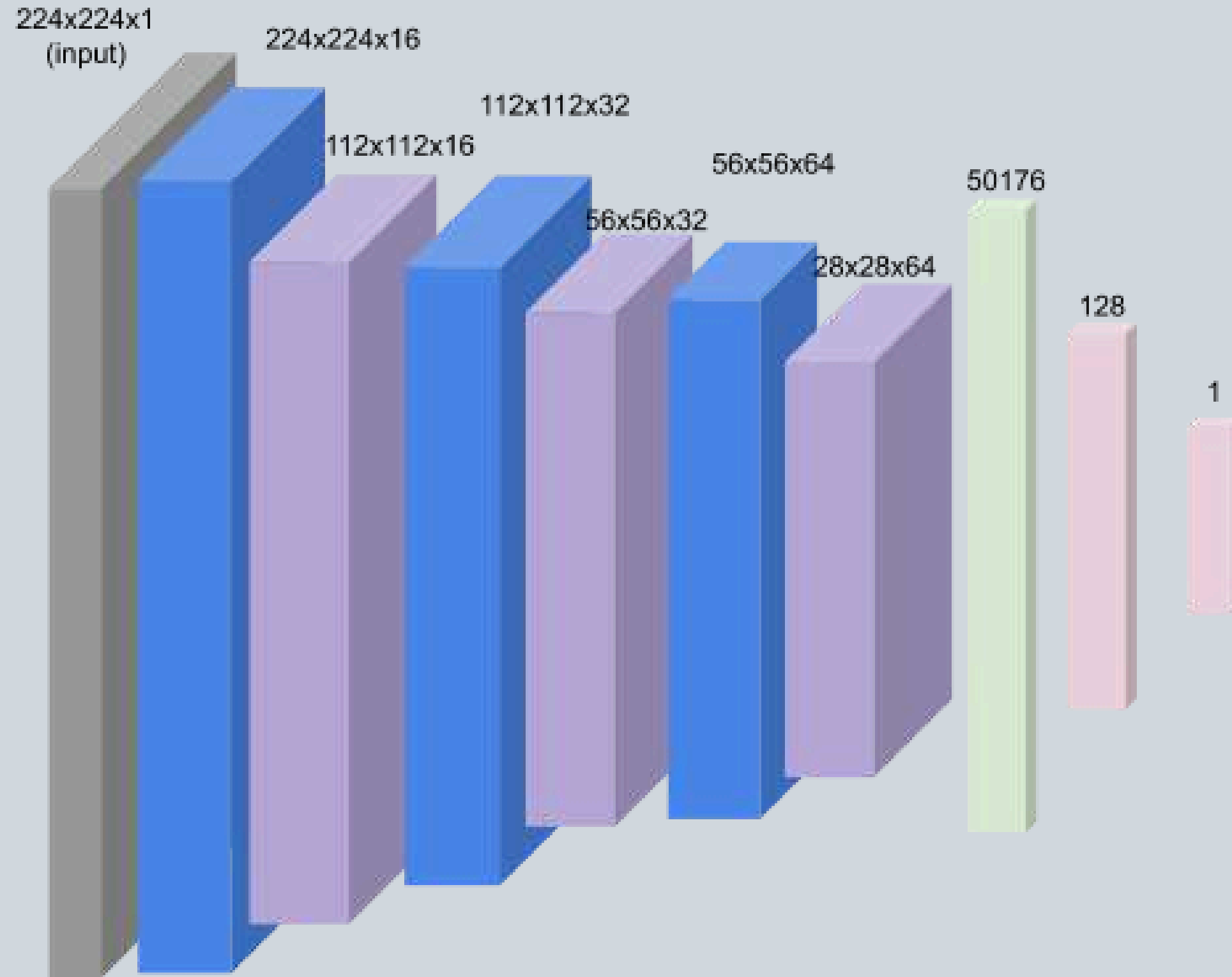
Rescales pixel (normalize) values from $[0, 1]$ to $[-1, 1]$ → centers around zero, converge faster

Convert **rayscale**



FROM BASELINE TO EFFICIENT MODEL

Baseline model



Data augmentation

- random horizontal flip
- random rotation ($\pm 10^\circ$)
- random brightness and contrast

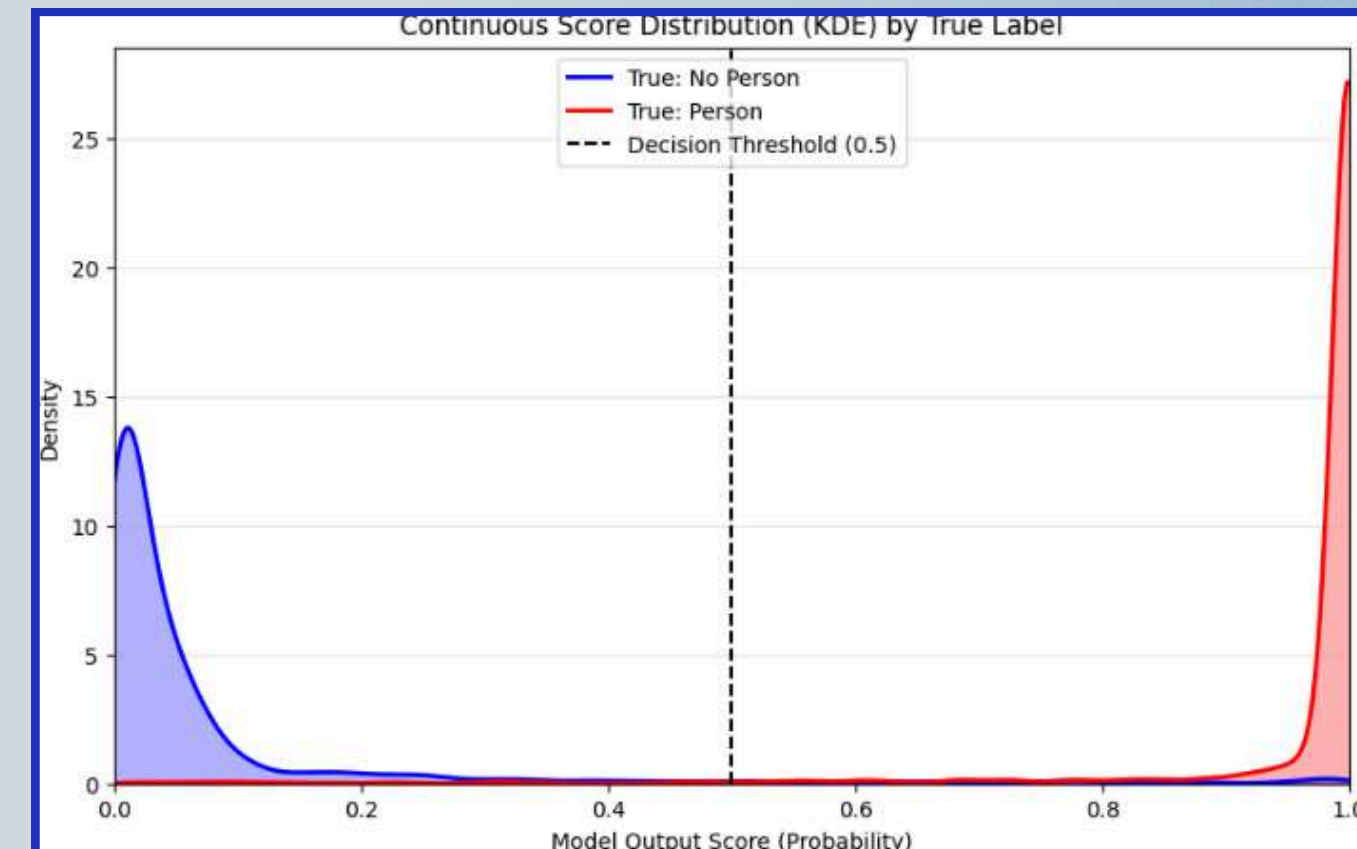
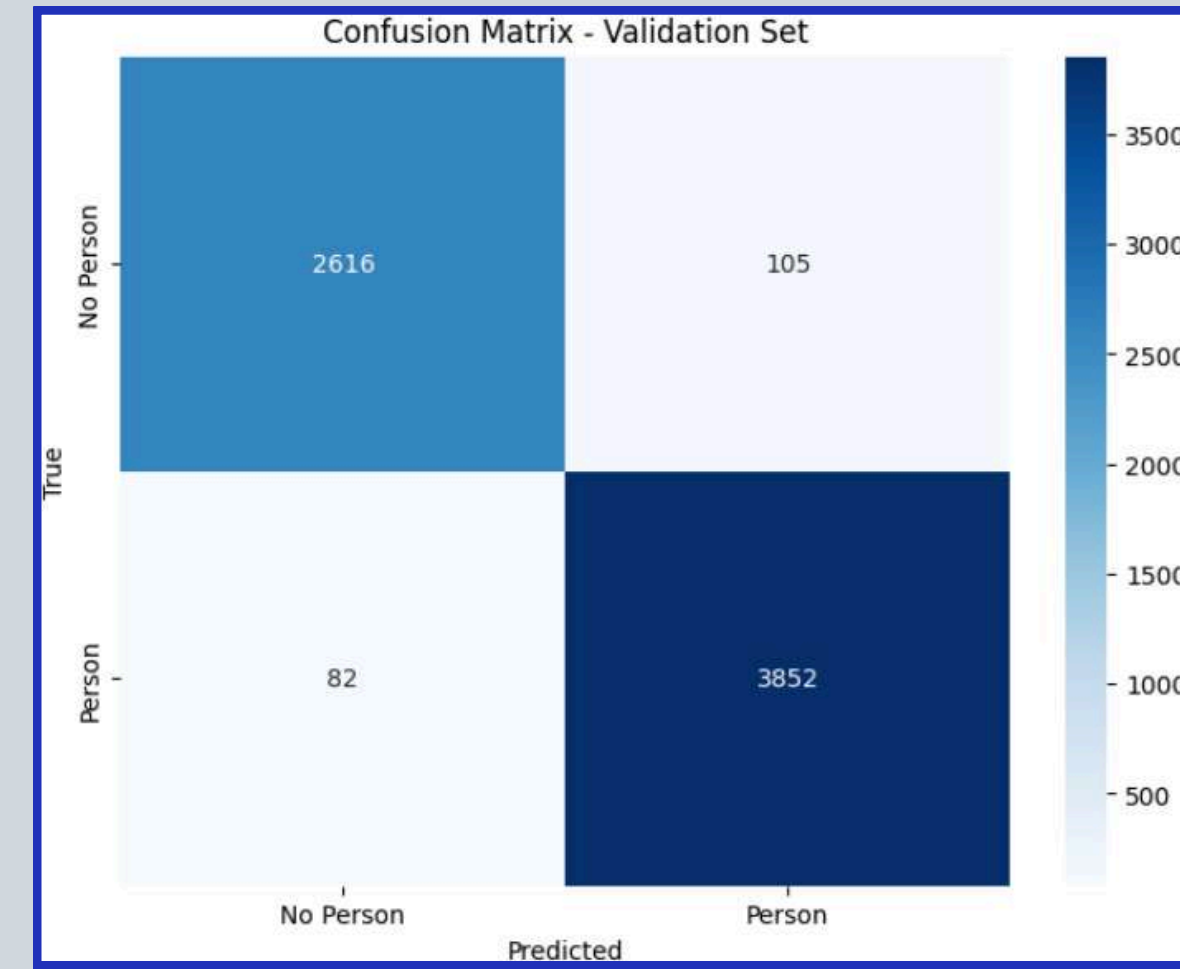
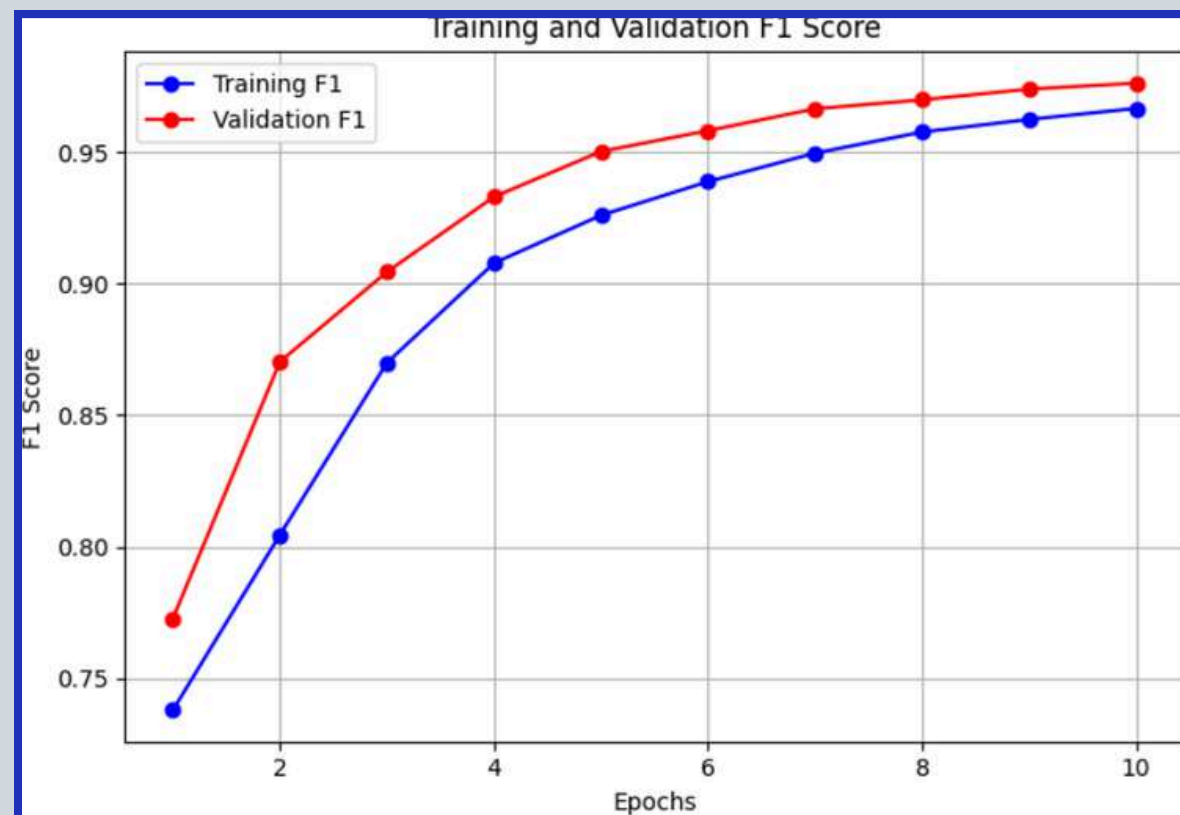
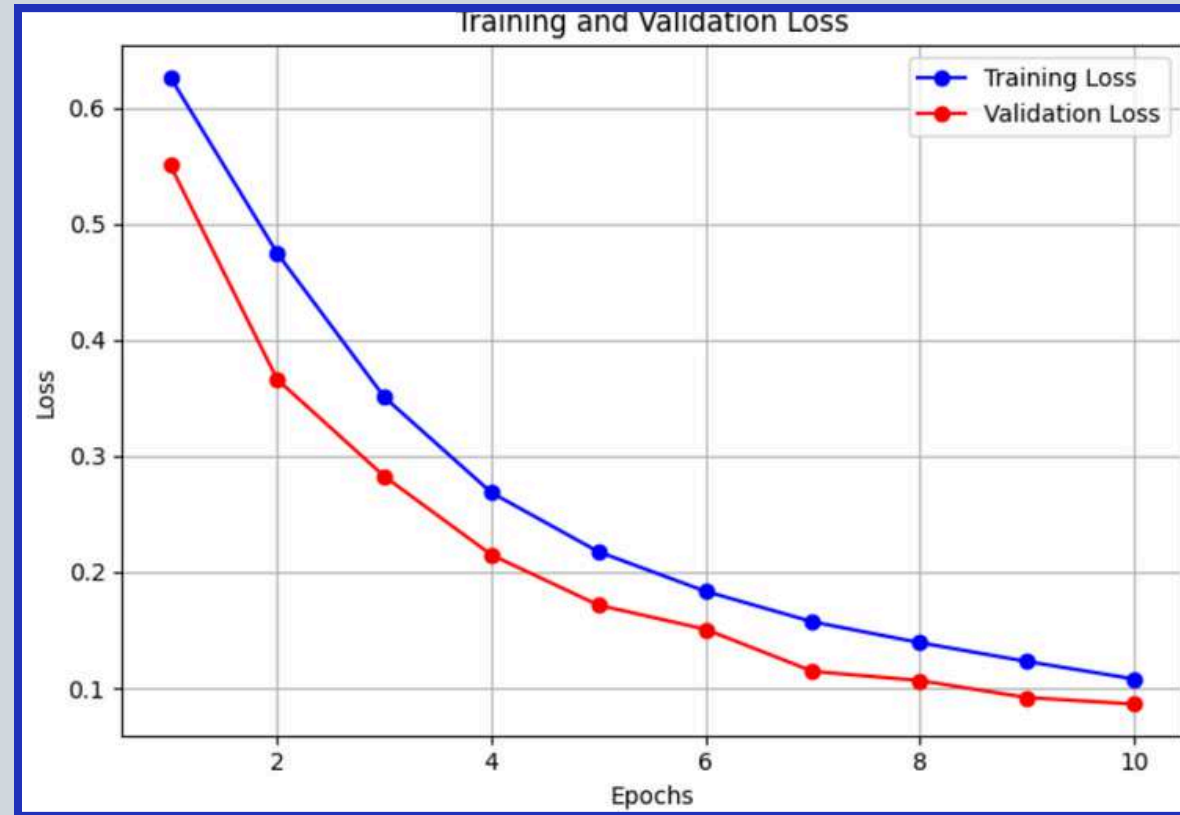
Loss function: Binary Cross Entropy loss

Optimizer: SGD with momentum



Total params: ~6,422,785

Baseline model



Experiment

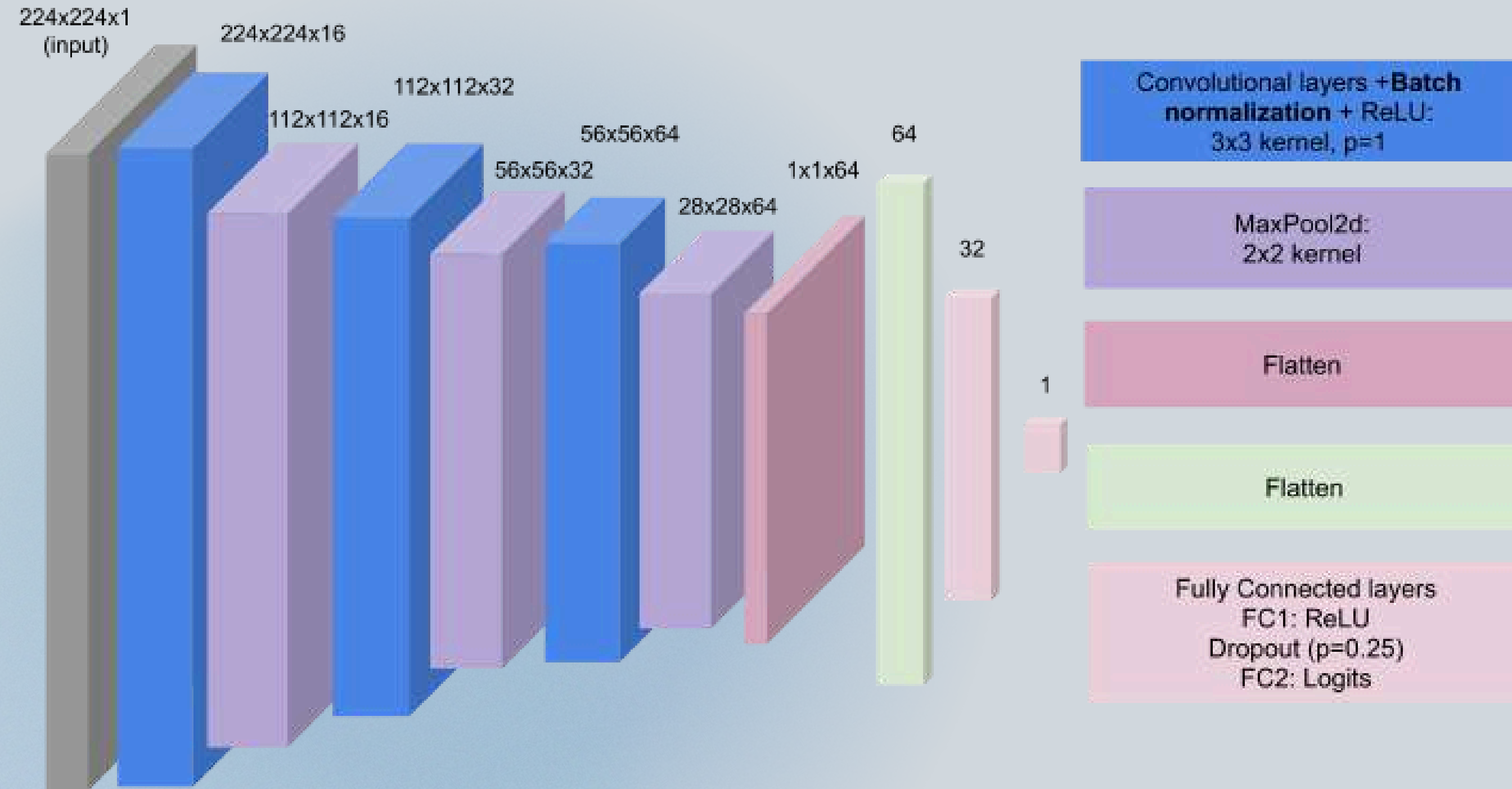
The experiments aimed to enhance a baseline model for a specific problem by:

- **Reducing parameters** from over **6.4M** to under **100,000** using **Adaptive Average Pooling** to address overfitting and performance.
 - Comparing the baseline's **SGD with momentum** to **Adam with Decoupled Weight Decay** as the optimizer.
 - Testing the output of probabilities versus logits by removing the last sigmoid activation and replacing **BCELoss** with **BCEWithLogitsLoss**.
 - Evaluating **Online Data Augmentation** against **Baseline's Data Augmentation** to improve generalization.
 - Assessing the effects of **batch normalization**.
- 

Results

	Train Loss	Train accuracy	Train F1	Validation Loss	Validation accuracy	Validation F1
AdapAvgPool	0.6768	0.5858	0.7388	0.6745	0.5911	0.7430
AdamW	0.2973	0.8771	0.8969	0.2318	0.9047	0.9232
Logits	0.2880	0.8843	0.9017	0.223	0.9082	0.9236
Better ODA	0.2307	0.9082	0.9222	0.1979	0.9235	0.9347
BatchNorm	0.1771	0.9296	0.9385	0.1380	0.9394	0.9470

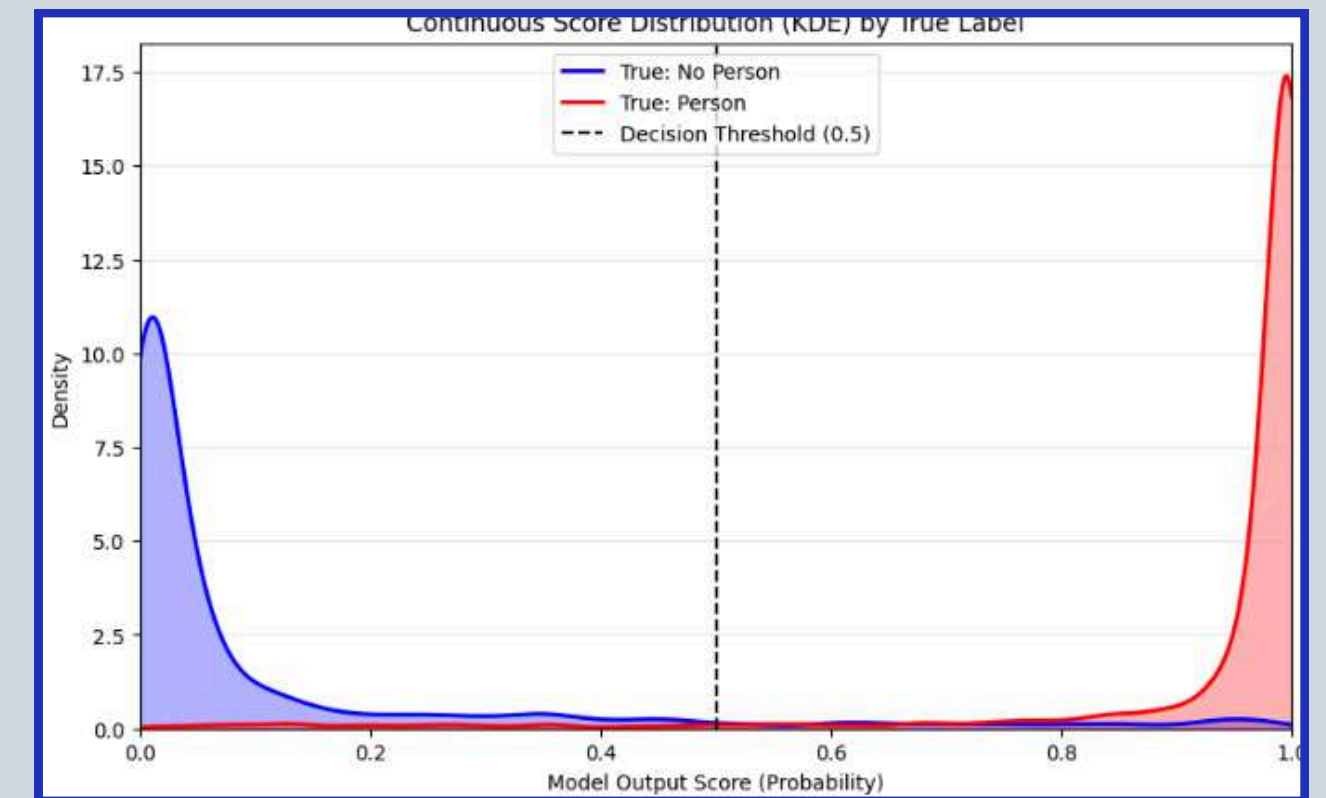
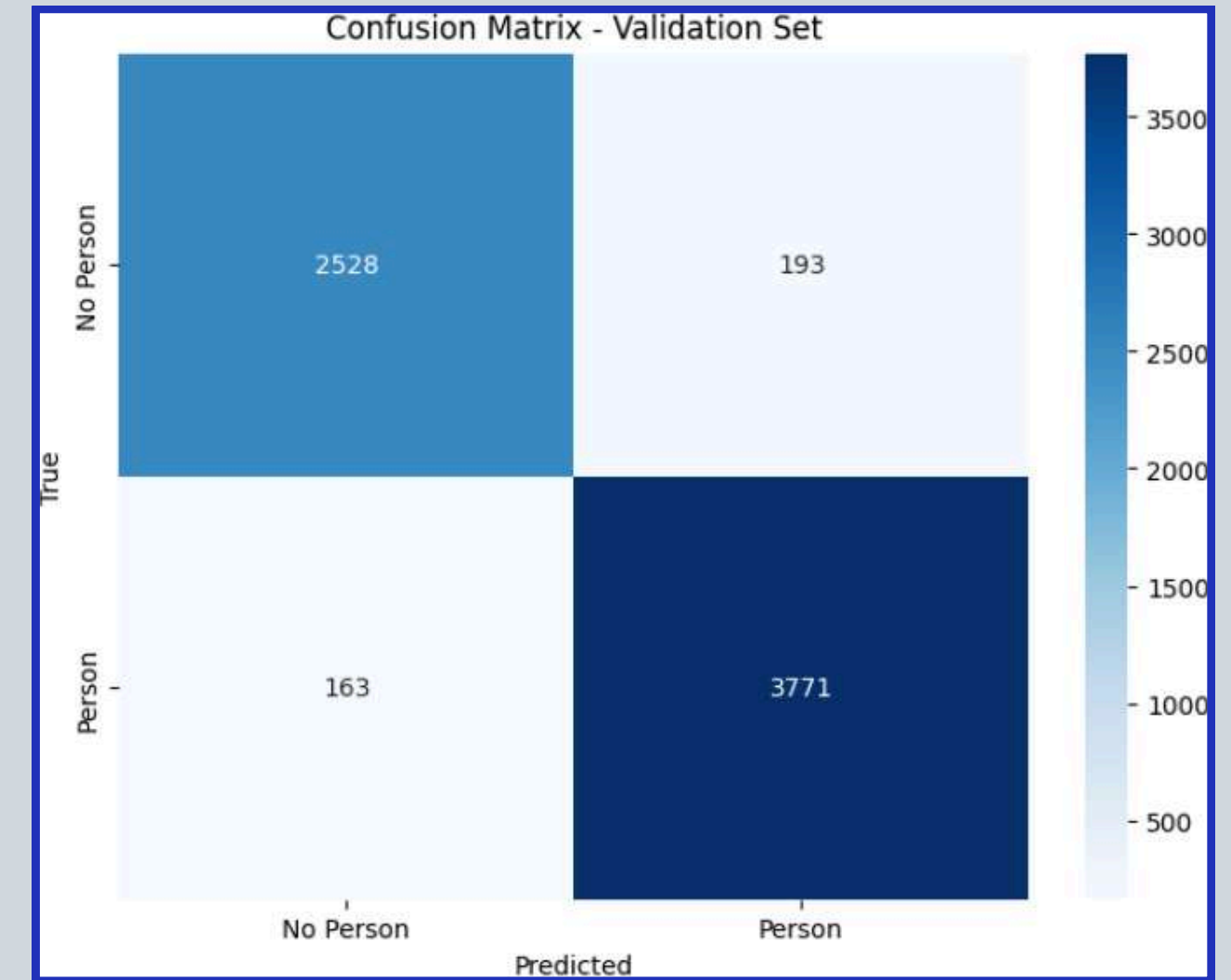
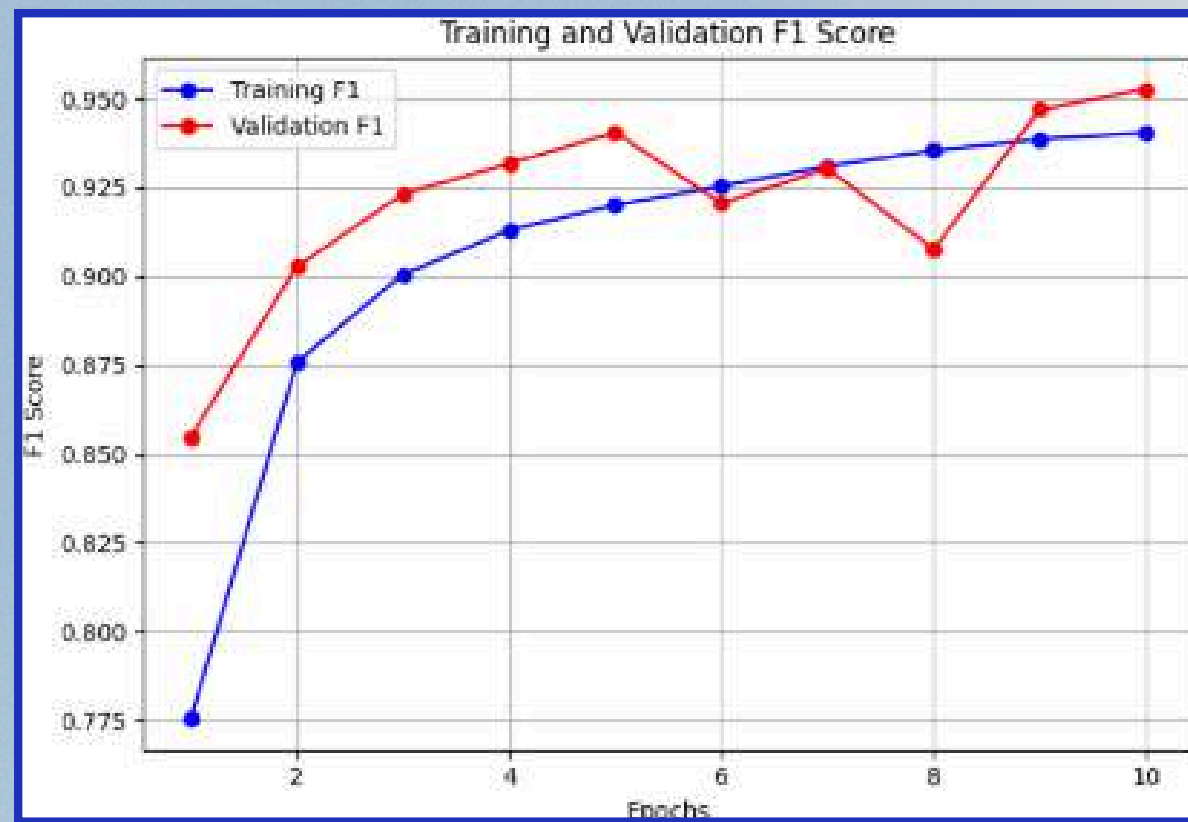
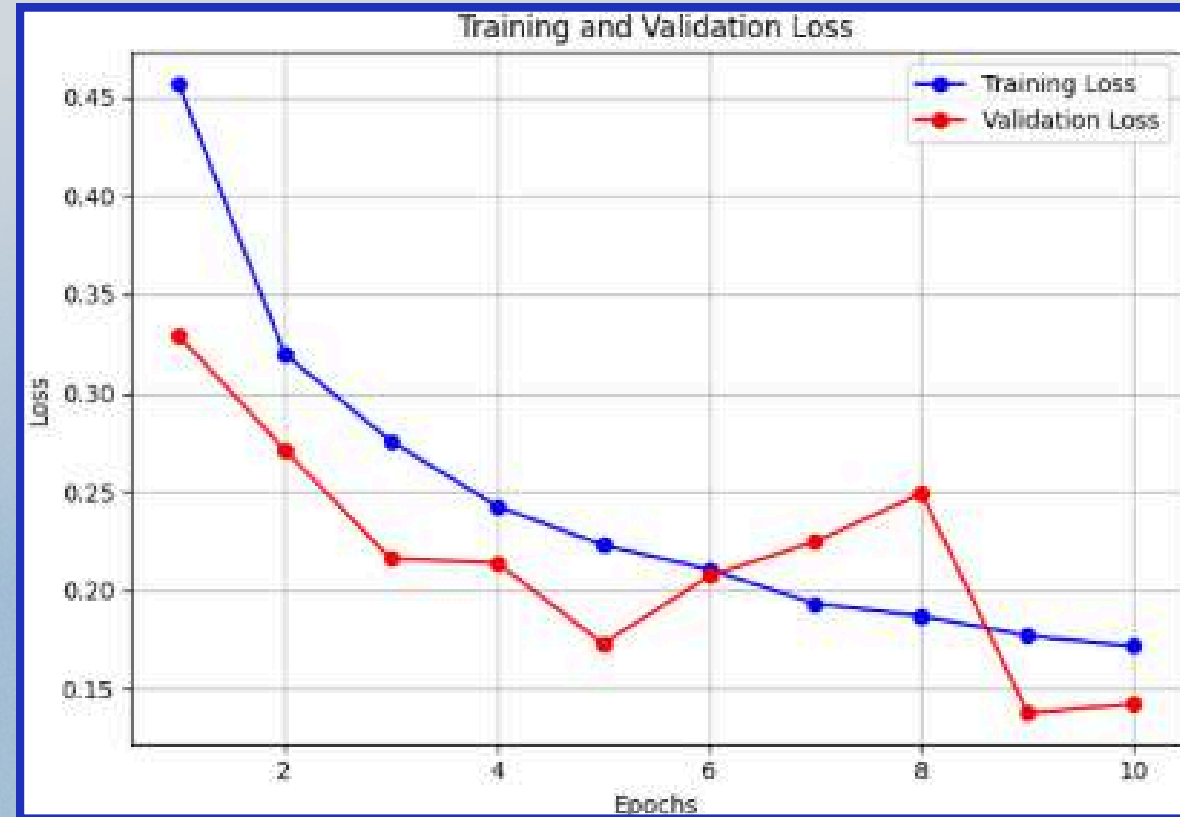
Results: final architecture



Data augmentation

- Random Horizontal Flip
- Random Rotation ($\pm 5^\circ$)
- Random Brightness and Contrast
- Random Affine Transformation
- Gaussian Blur
- Gaussian Noise

Results: final model



Analysis of the result

LLVIP test dataset

- Accuracy: 0.8182
- F1 Score: 0.8358

My dataset

- Accuracy: 0.3728
- F1 Score: 0.2536

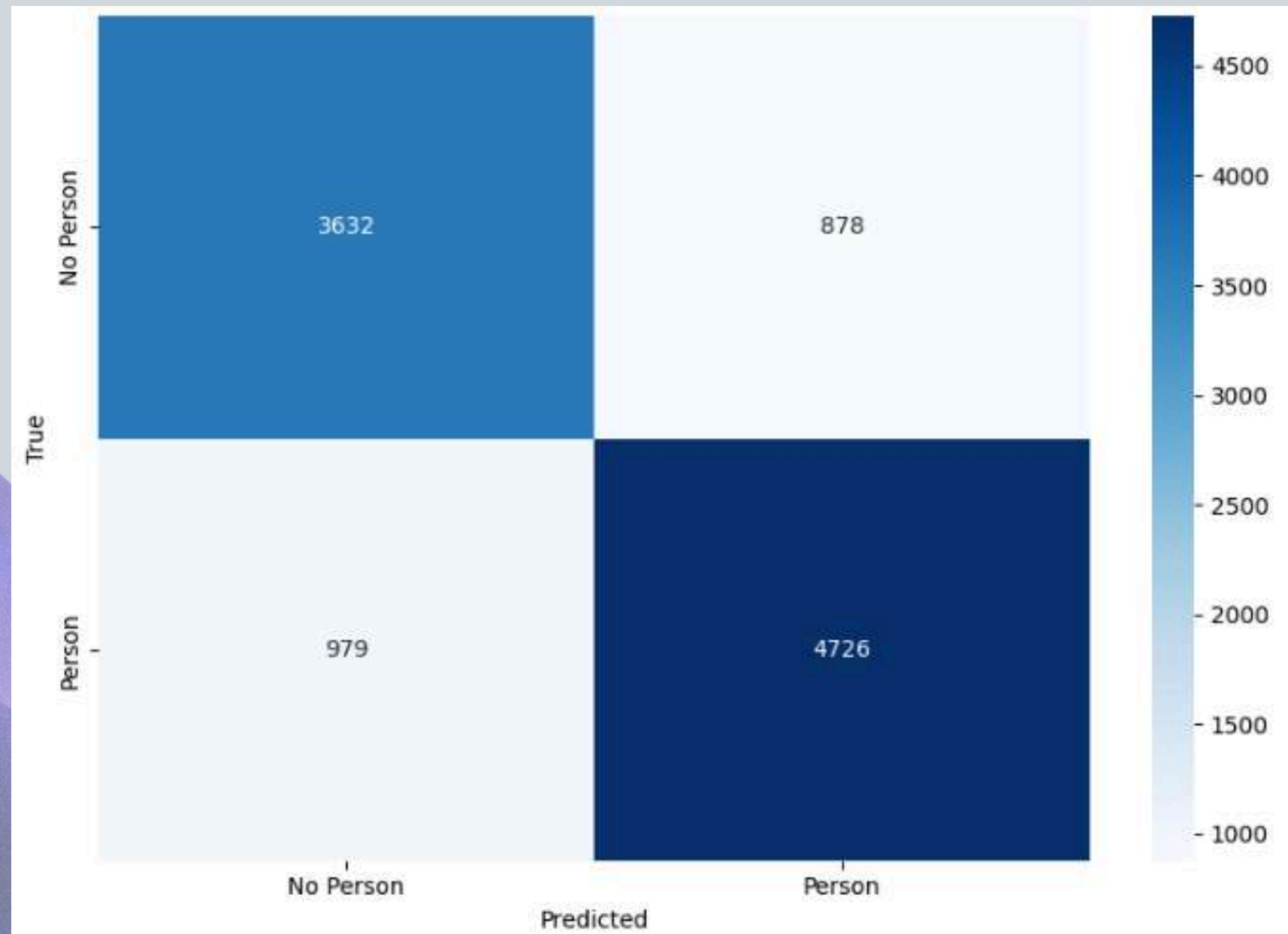
Domain shift

```
graph TD; A[LLVIP test dataset] --> C[Domain shift]; B[My dataset] --> C;
```

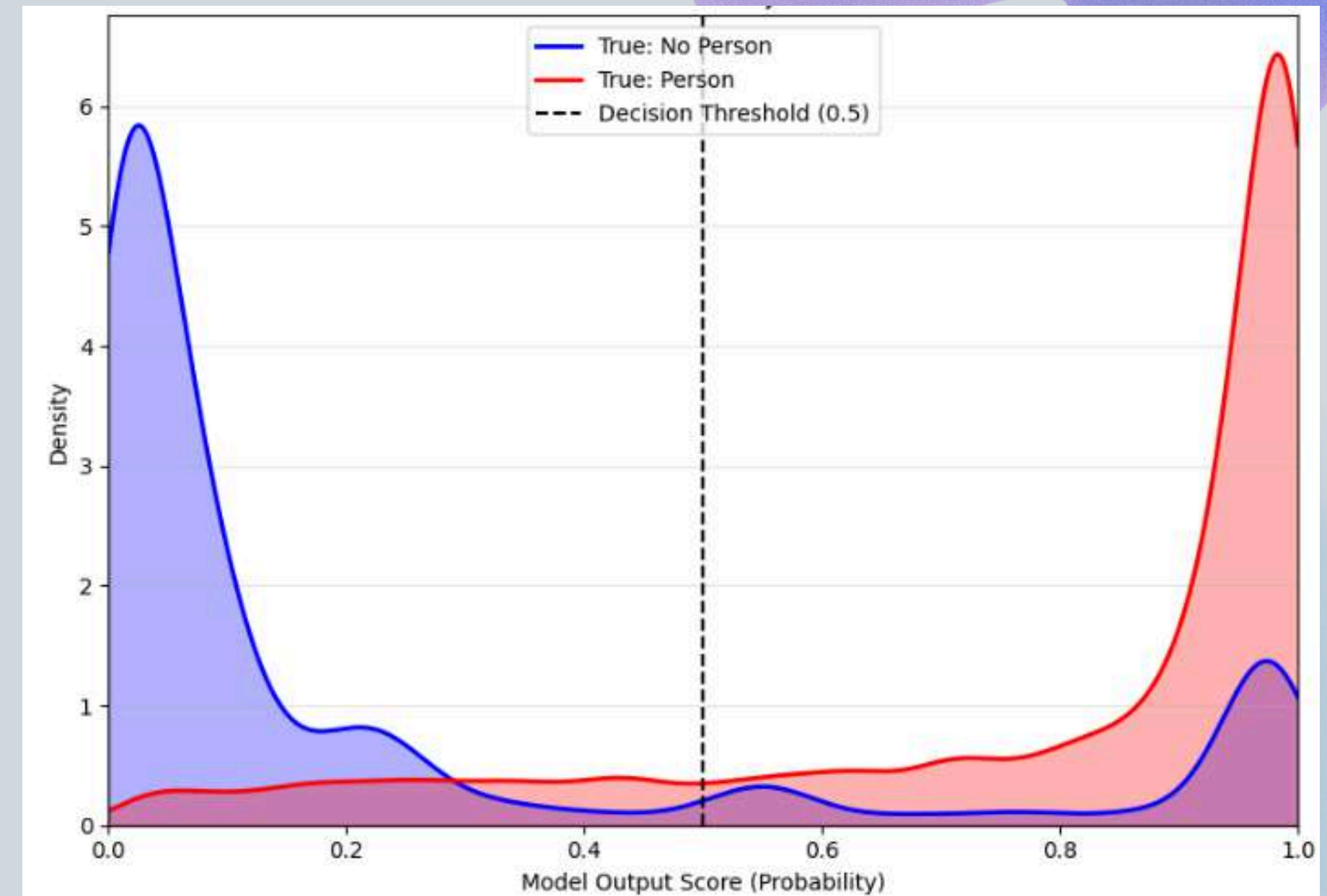
The diagram illustrates a domain shift between two datasets. Two rounded rectangular boxes at the top represent the 'LLVIP test dataset' and 'My dataset'. The LLVIP dataset box contains the text '- Accuracy: 0.8182' and '- F1 Score: 0.8358'. The 'My dataset' box contains '- Accuracy: 0.3728' and '- F1 Score: 0.2536'. Two purple arrows originate from the bottom of each box and point towards a central text label 'Domain shift' at the bottom center of the slide. The background features abstract purple and blue organic shapes.

Analysis of the result

Confusion matrix on test split: LLVIP

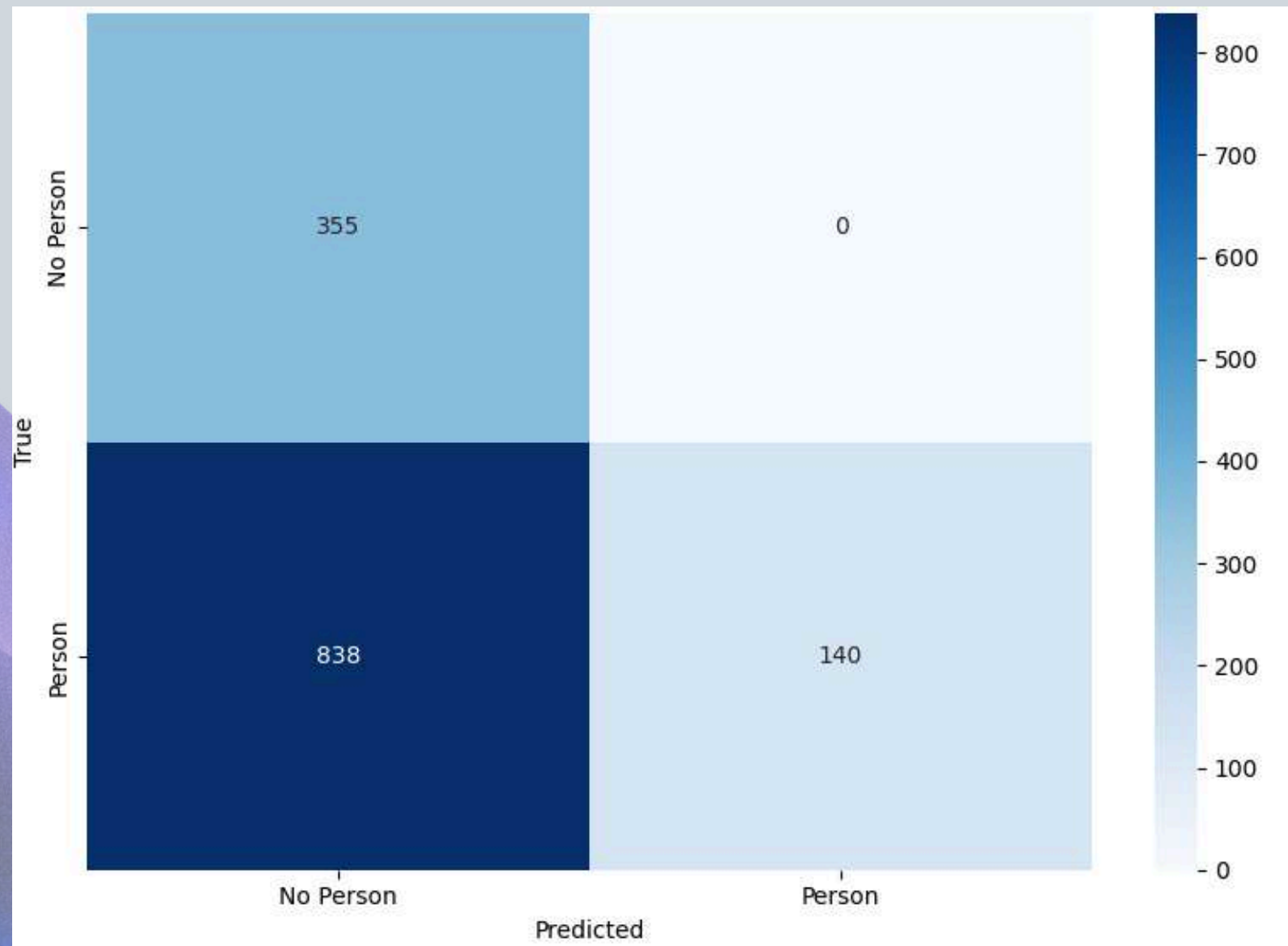


Continuos score distribution: LLVIP

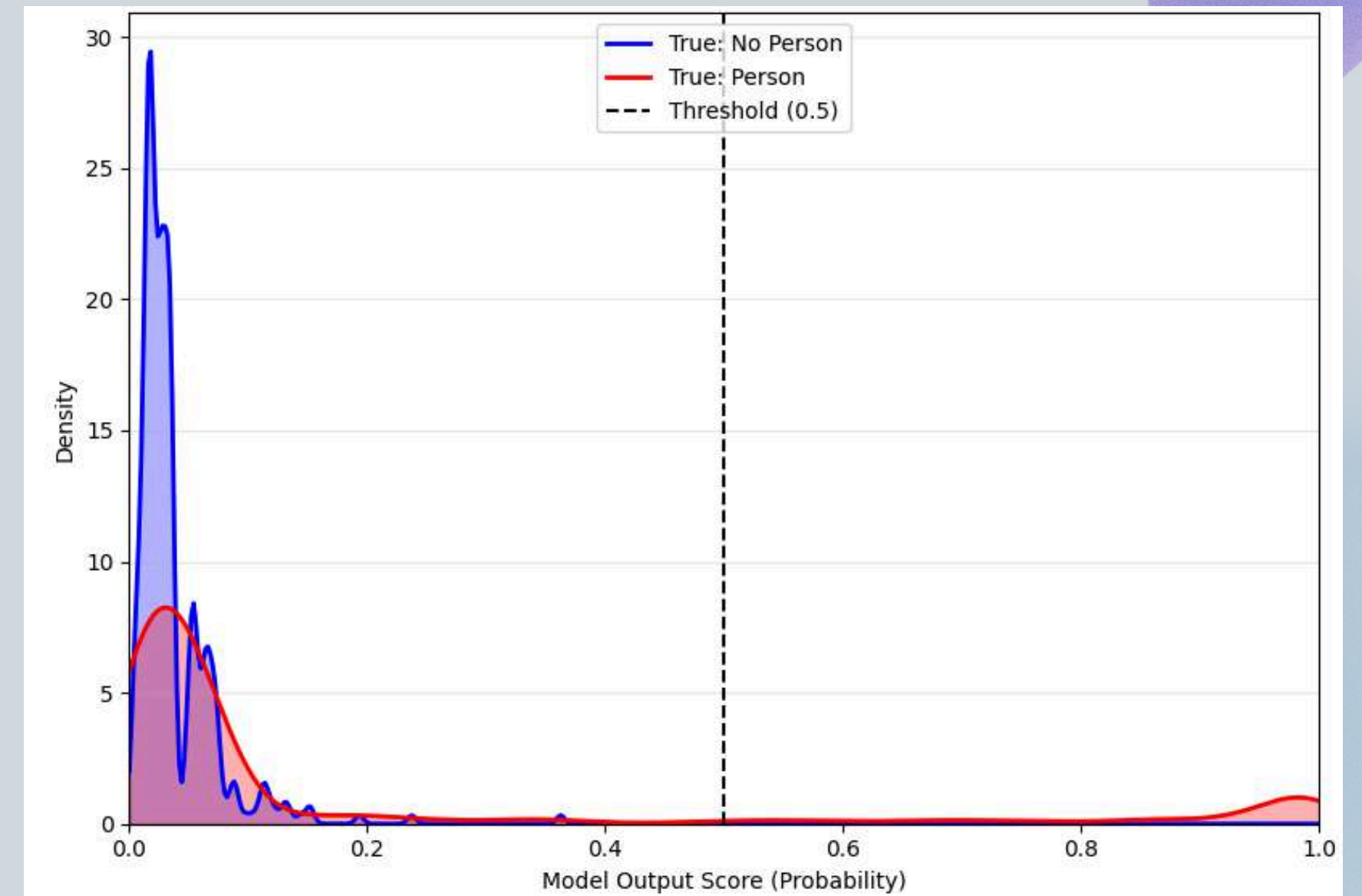


Analysis of the result

Confusion matrix on test split: My dataset

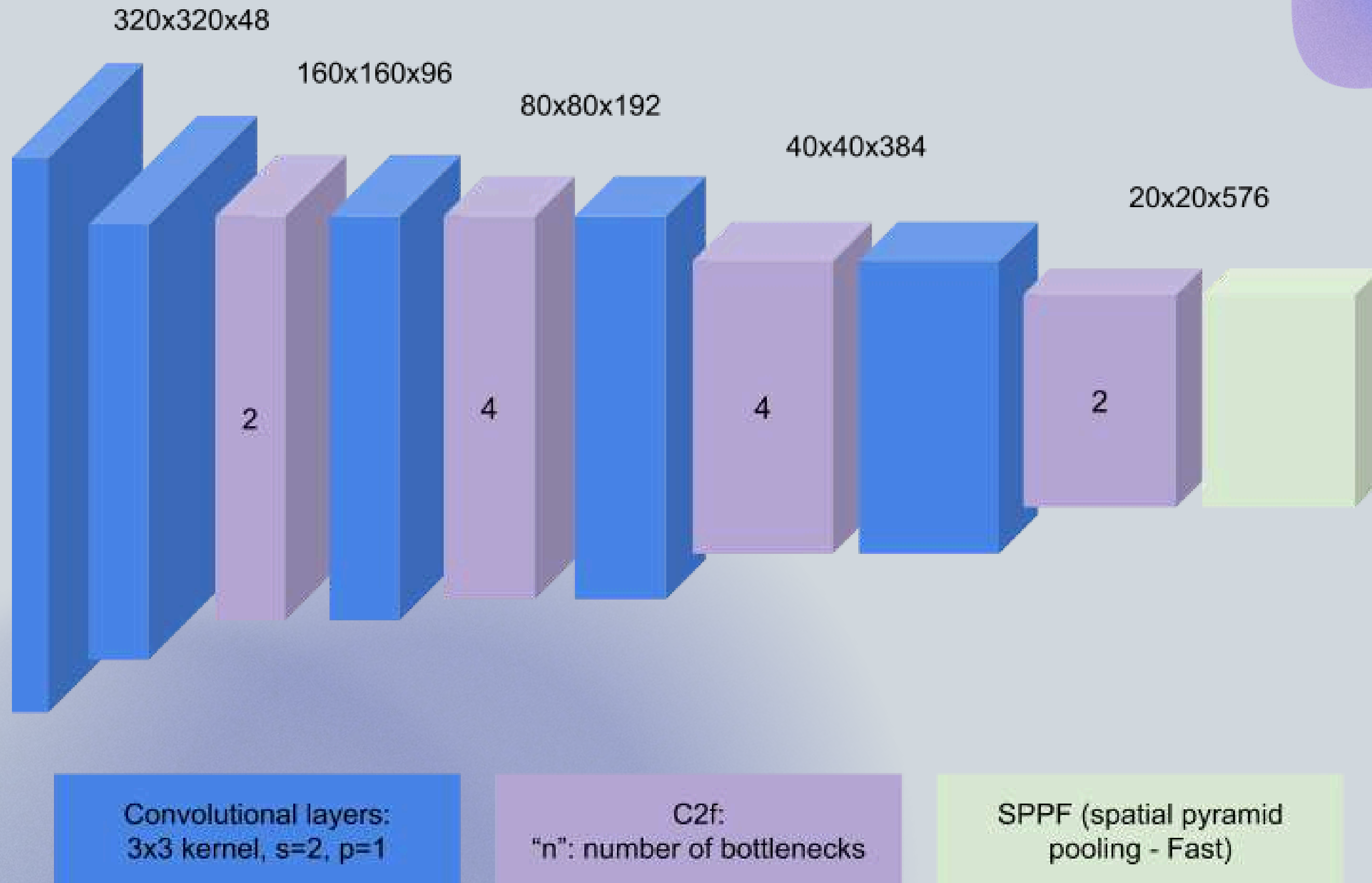


Continuos score distribution: My dataset

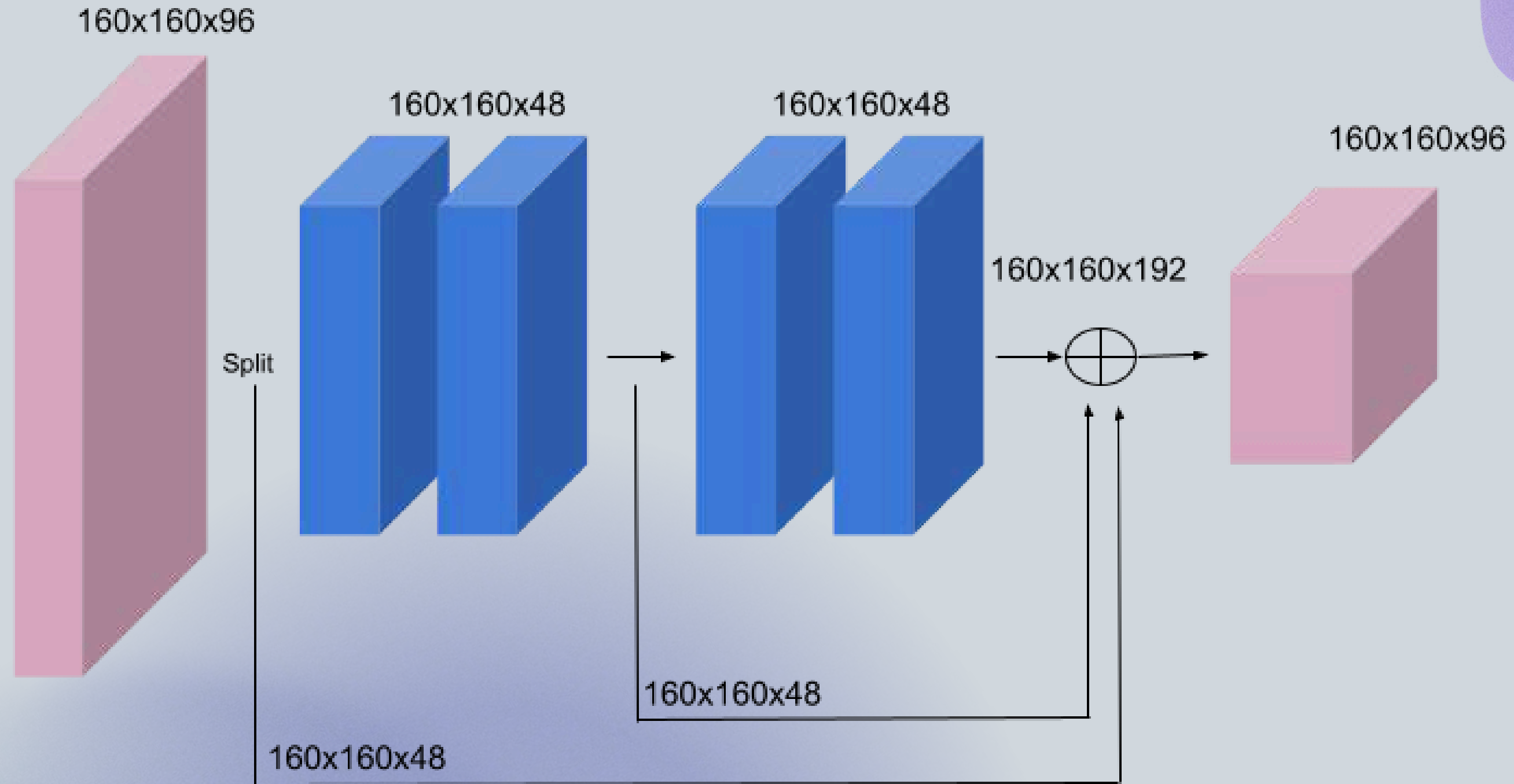


TRANSFER LEARNING

YOLOv8 backbone



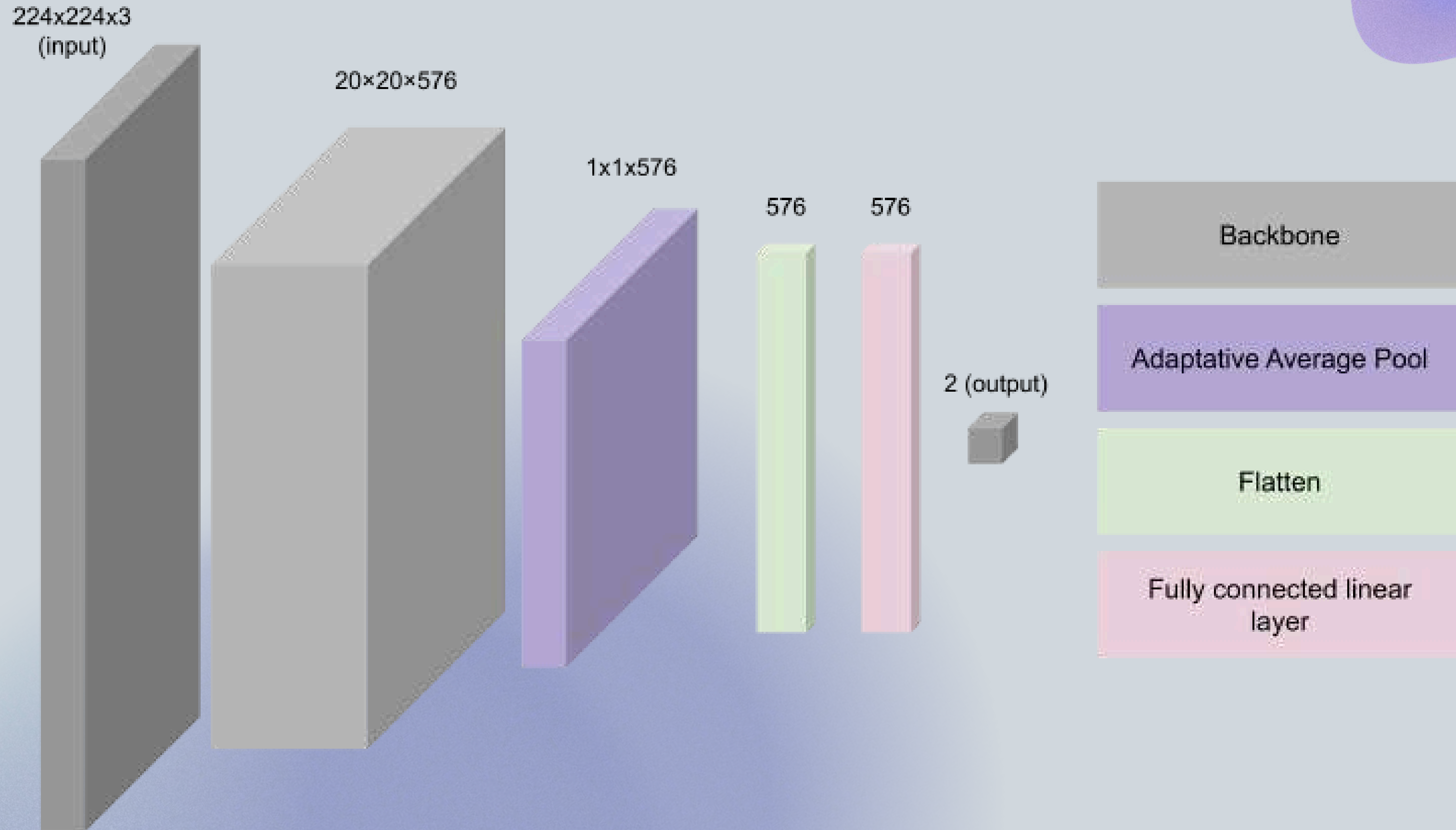
C2f (Concatenate to Feature)



Convolutional layer:
1x1 kernel, s=1, p=1

Convolutional layers:
3x3 kernel, s=1, p=1

Transfer learning model



Experiment

- Cross Entropy Loss
- Adam: learning rate 0.001
- 5 epochs
- Split: 70% training, 15% validation and 15% testing

PREPROCESSING

Applied to all splits: train, val, and test

Resize 224×224

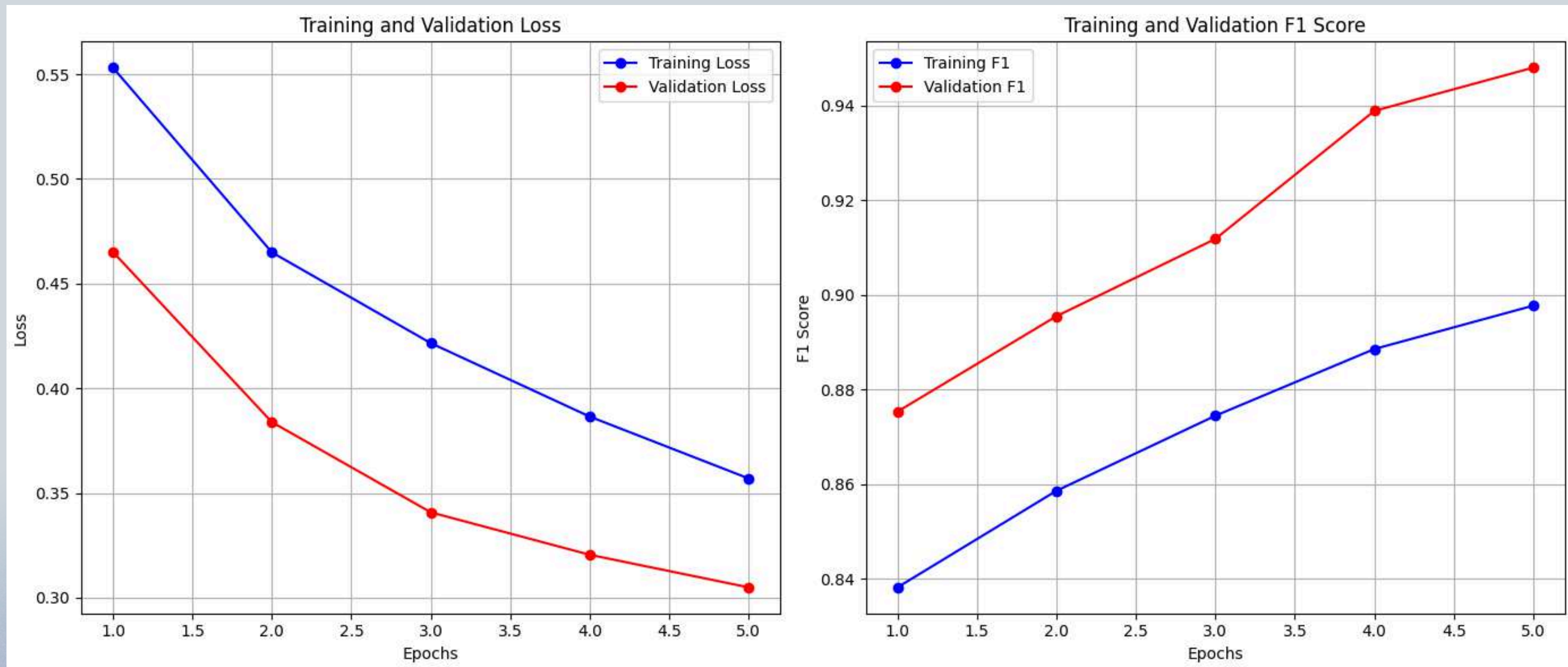
Converts Image to a PyTorch tensor

Rescales pixel (normalize):

mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]

Results

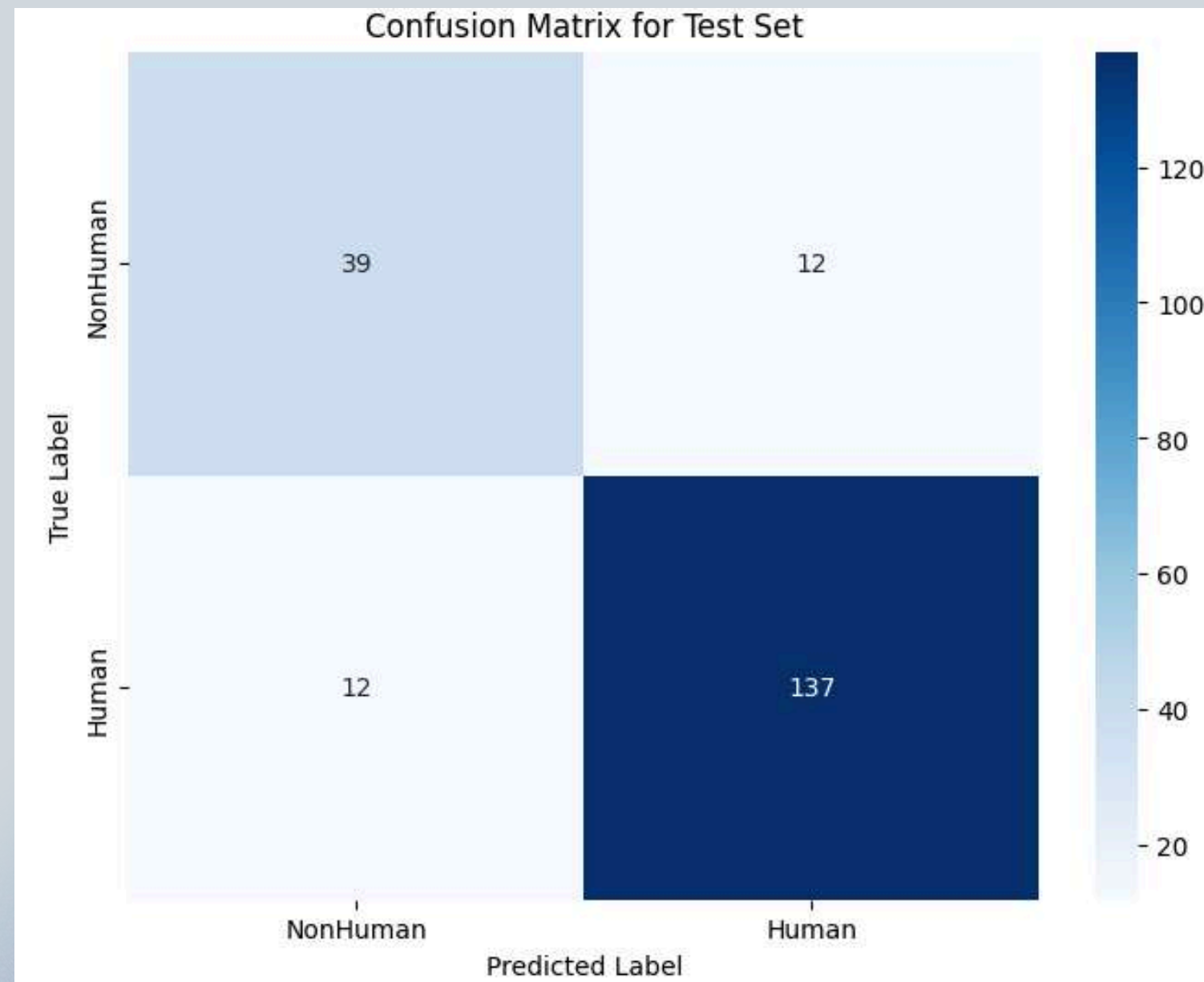
- Final Validation Accuracy: 90.00%
- Final Validation F1 Score: 0.9471



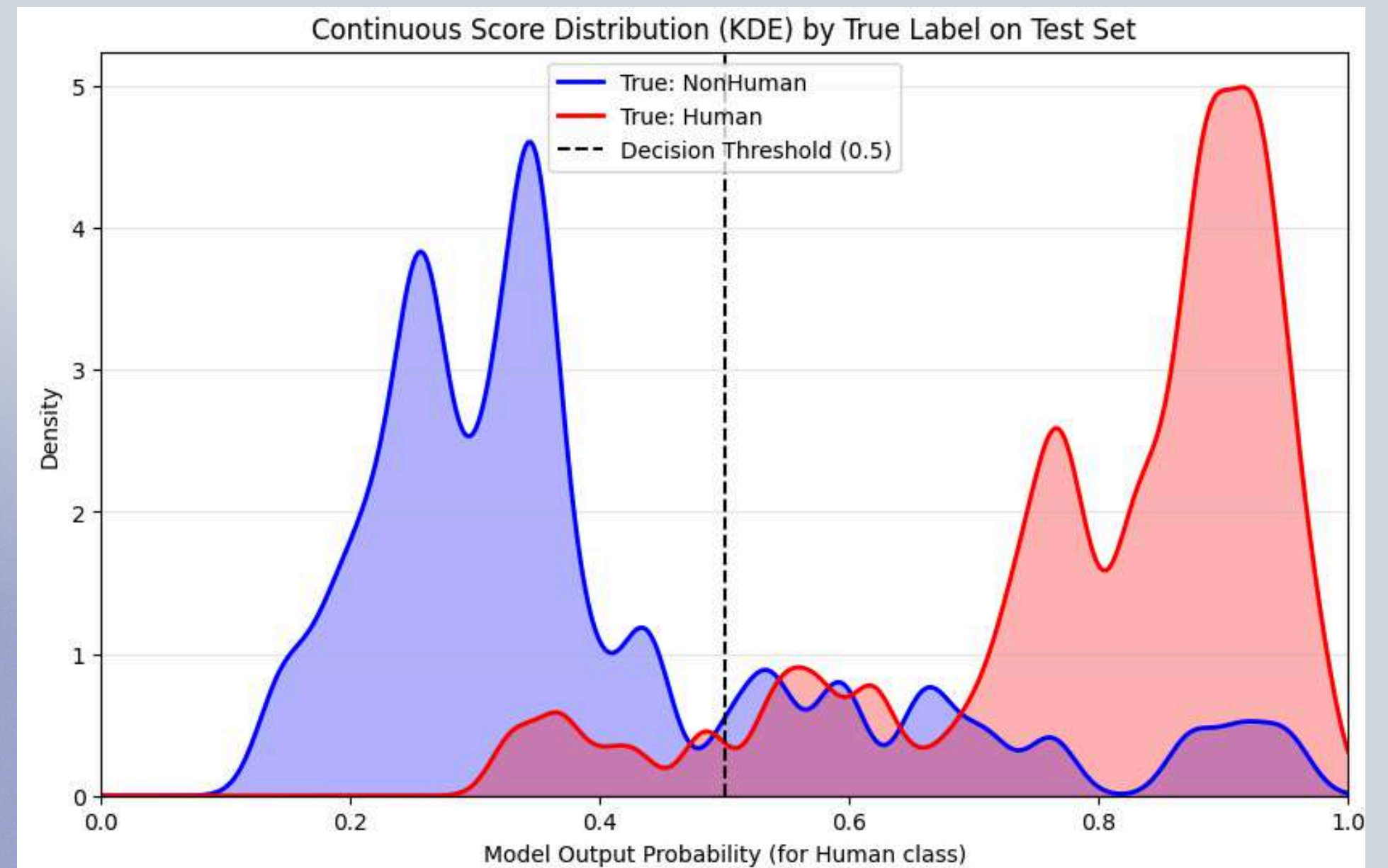
Results on test

- Final Test Accuracy: 89.50%
- Final Test F1 Score: 0.9307

Confusion matrix on test split



Continuous score distribution



LIMITATIONS

Domain Shift

Dramatic performance drop.

The visual characteristics differ significantly: thermal signatures vs. night images.

Difficult Cases

Person images with small std are nearly indistinguishable

Small Custom Dataset

Only 1333 images in my dataset → limited generalization capacity

Imbalanced and some similar images

Transfer Learning Normalization Mismatch

Infrared images are grayscale replicated to 3 channels, statistics differ significantly from natural images

FUTURE IMPROVEMENTS

- Collect more real "no person" frames for my_dataset to fix class imbalance. Collect more variate images with different backgrounds and conditions.
- Proper domain adaptation: fine-tune on my_dataset after training on LLVIP, or use domain randomization
- Explore thermal-specific pretrained models to do transfer learning
- Real-time deployment: integrate model into a live camera pipeline with automatic alert system
- Detection vs. classification: explore object detection for more precise localization of intruders

CONCLUSIONS

- Successfully trained a binary classifier for nighttime person detection in infrared images.
- Image variety is critical to obtain general models
- Transfer learning helps getting good results with not much training data
- Although great part of the transfer learning model is trained with daylight images, it is able to adapt to the new domain
- Domain shift remains a core challenge
- Batch normalization, AdamW, and logits-based loss are key design choices for this task



THANK YOU